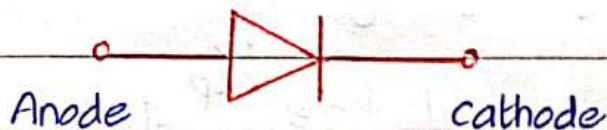
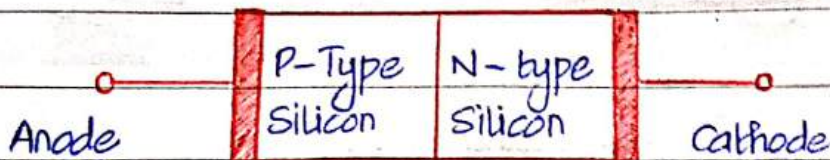


MODULE - 3

PN JUNCTION

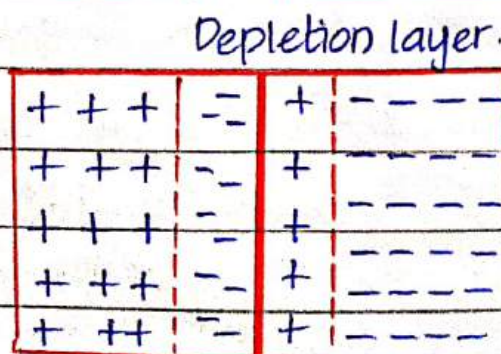
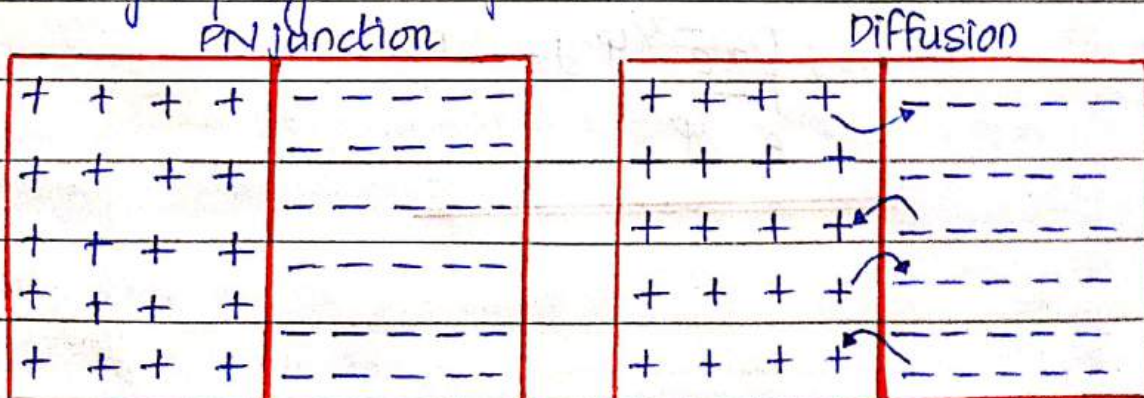


There is another type of diode called schottky diode. In schottky diodes, instead of P and N-type material, one side will be the metal side & the other side will be a semiconductor side.

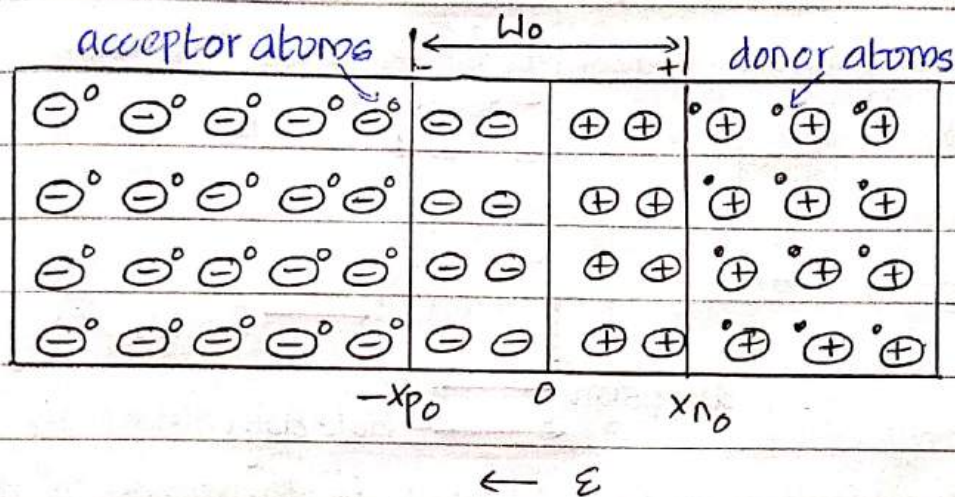


→ Schottky diodes - different categories (2 categories): ohmic contact and rectifying contact.

→ Note: Refer p-n junction formation.



Concentration gradient $\rightarrow$ movt of majority carriers (diffusion)
 $\rightarrow$ recombination near to the junction / at the junction.



Ionization occurs $\rightarrow$ ionized charges at both the sides $\rightarrow$ due to this charge, creation of an electric field $\rightarrow$ drifting of charge carriers.

Note:—Initially, there is diffusion and space charge is created, then electric field is created. When the electric field is created, there is a drift of charge carriers.

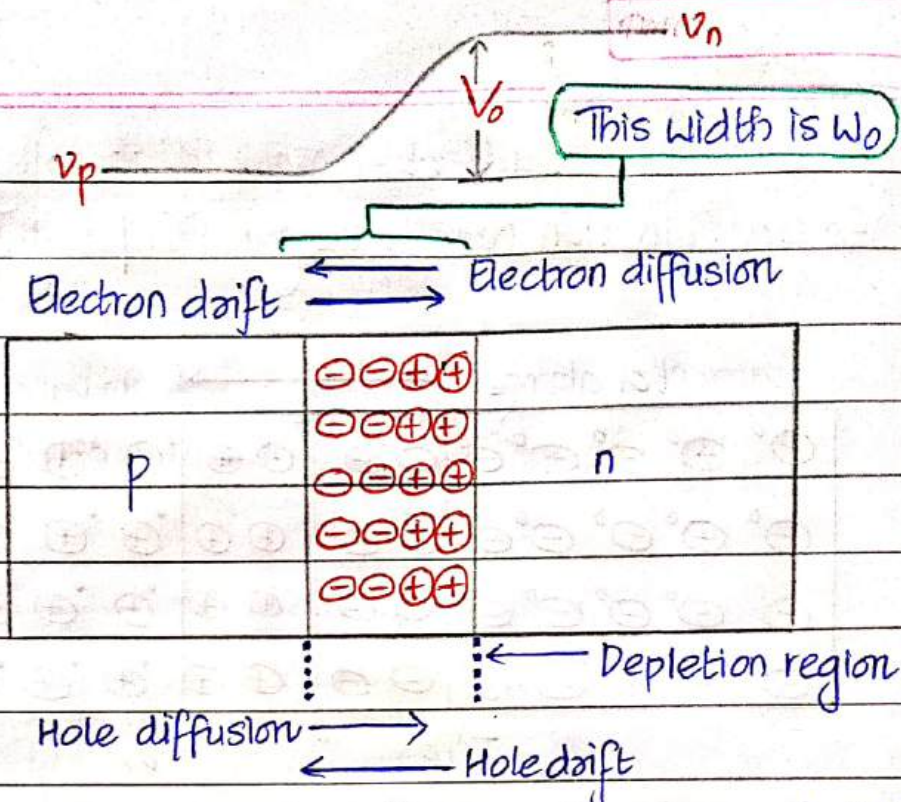
$\rightarrow$ When the space charge increases, diffusion decreases. Whereas due to the creation of the electric field, the drift current will be increased.

$\rightarrow$ So as diffusion $\downarrow$, electric field $\uparrow$ and at a stage, there will be a balance i.e. diffusion and drift current gets balanced. That particular point is known as the **equilibrium point**.

At the eqbm point; the current due to diffusion balances the current due to drift, and their sum will be 0.

This is the reason why there is no net current under eqbm condition.

- Depletion region is also called transition | space charge region



- When the junction is created, there is the process of diffusion & drift until eqbm.
- Movement of charge carriers: electron diff :- $n \rightarrow p$ side
hole diff :- $p \rightarrow n$ side

Electric field from +ve to -ve. Hole moves in the direction of electric field and e^- s drift opposite to its direction.

- e^- s present in the p side are minority carriers; also the holes in the n side. $\Rightarrow$ drift action is due to the minority carriers and diffusion due to majority carriers.

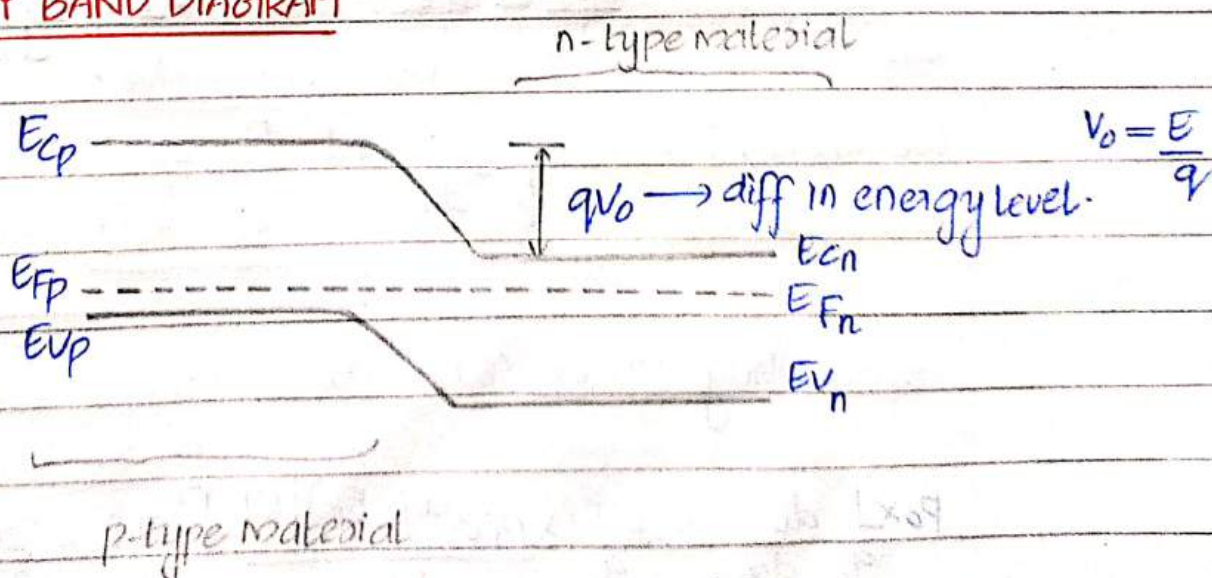
Under eqbm condition,

$$J_n = J_n(\text{diff}) + J_n(\text{drift}) = 0$$

$$J_p = J_p(\text{diff}) + J_p(\text{drift}) = 0$$

- Since there is an electric field, there will ^{be} an electrostatic potential and that potential difference is represented by V_0 . It is the potential difference across the junction.
- At the n side, there is positive charge while negative charge is present at the p side. Hence potential at n side is at a higher state (V_n) and potential at the p side is at a lower state (V_p). The difference between these two potentials is the **contact potential (V_0)/barrier potential**.
- If we measure V_0 across the junction, the potential will be equal to 0 because new contact potential will be formed at each probe, and that potential cancels the potential, V_0 .
Hence, it cannot be measured using a voltmeter (value will be 0).
- V_0 is the potential created under the eqbm condition; it will not produce any net current flow.

ENERGY BAND DIAGRAM



→ In the case of p-side/p-type, the fermi level lies closer to the valence band and in the n-type material, fermi level lies closer to the conduction band.

→ When the junction is formed, fermi level will maintain a constant energy level (It will be same throughout the material even if the junction is formed).

CONSTANCY OF FERMİ LEVEL UNDER EQUILIBRIUM

Under equilibrium, $J_p = J_n = 0$

$$J_p = q p_0 \mu_p E(x) - q D_p \frac{dp_0}{dx} = 0$$

$$q p_0 \mu_p E(x) = q D_p \frac{dp_0}{dx}$$

$$p_0 E(x) = \frac{D_p}{\mu_p} \frac{dp_0}{dx} \quad \text{--- (1)}$$

$$p_0 = n_i e^{(E_i - E_F)/KT}$$

$$\frac{dp_0}{dx} = n_i e^{(E_i - E_F)/KT} \times \frac{1}{KT} \left[\frac{dE_i}{dx} - \frac{dE_F}{dx} \right] \quad \text{--- (2)}$$

$$\text{We know, } E(x) = \frac{1}{q} \frac{dE_i}{dx} \text{ and } \frac{D_p}{\mu_p} = \frac{kT}{q}$$

$$\text{--- (3)}$$

Substituting (2) and (3) in (1),

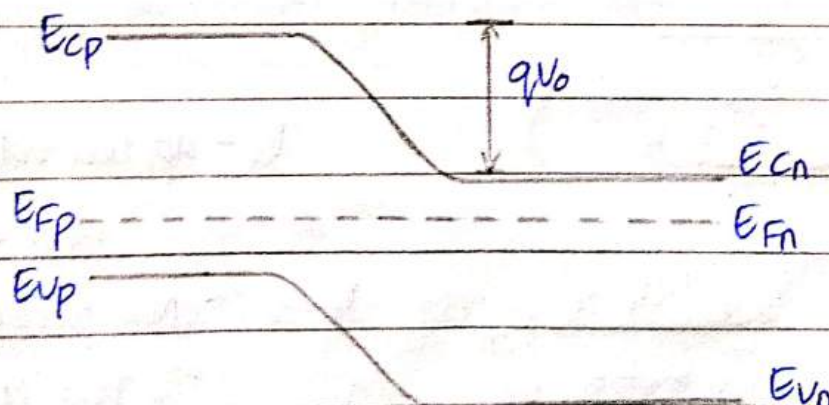
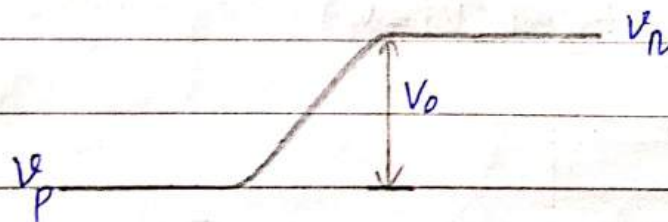
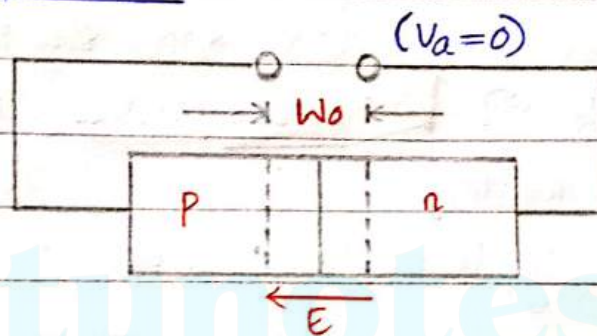
$$p_0 \times \frac{1}{q} \frac{dE_i}{dx} = \frac{kT}{q} \times \frac{n_i e^{(E_i - E_F)/KT}}{KT} \left[\frac{dE_i}{dx} - \frac{dE_F}{dx} \right]$$

$$\frac{p_0}{q} \frac{dE_i}{dx} = \frac{p_0}{q} \left[\frac{dE_i}{dx} - \frac{dE_F}{dx} \right]$$

$$\Rightarrow \underline{\underline{\frac{dE_F}{dx} = 0}}$$

ENERGY BAND DIAGRAM AND CURRENT FLOW

(i) Under equilibrium,



PARTICLE FLOW	CURRENT FLOW
(1) $\longrightarrow$	$\longrightarrow$
(2) $\longleftarrow$	$\longleftarrow$
(3) $\dashleftarrow$	$\dashrightarrow$
(4) $\dashrightarrow$	$\dashrightarrow$

(1) Hole diffusion

(2) Hole drift

(3) Electron diffusion

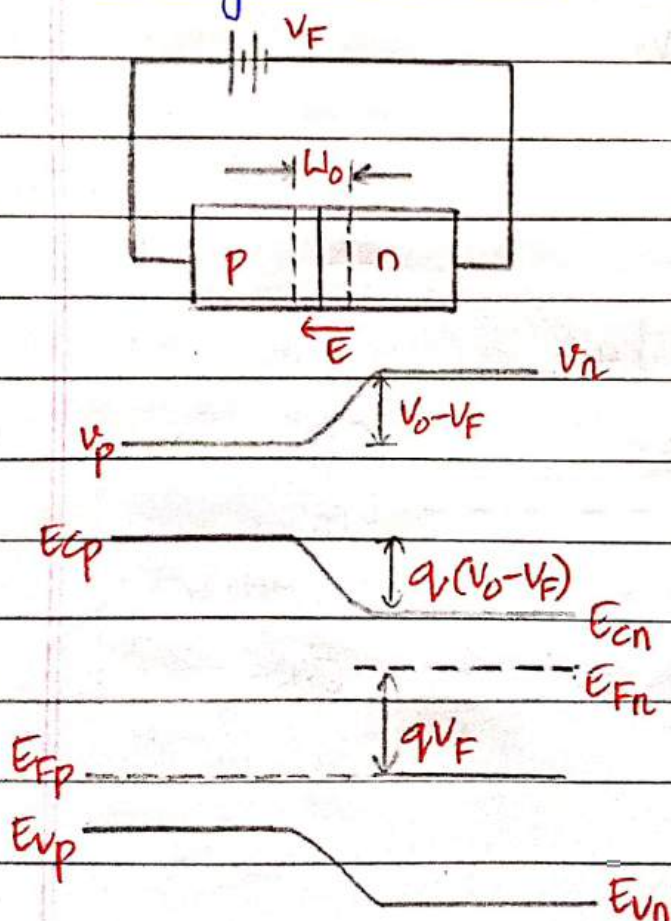
(4) Electron drift

The size of the arrows shows the amount of current.

(almost same size here $\Rightarrow$ also these values balance each other)
 hole diffusion, hole drift
 electron diffusion and drift

$\rightarrow$ After maintaining this balance, we obtain width W_0 and electric field E .

(ii) When forward biased ($V_a = V_F$)



V_a - applied voltage (+ve for FB)
and -ve for R.B

$V_0 - V_a$ - net potential at the barrier.

PARTICLE FLOW	CURRENT FLOW
(1) $\longrightarrow$	$\longrightarrow$
(2) $\longleftarrow$	$\longleftarrow$
(3) $\longleftarrow \text{---}$	$\text{---} \longrightarrow$
(4) $\text{---} \longrightarrow$	$\longleftarrow \text{---}$

(1) Hole diffusion

(2) Hole drift

(3) Electron diffusion

(4) Electron drift

Size indicates diffusion is more

→ In forward biased, p-side to +ve terminal and n-side to -ve terminal of the battery.

→ Assume that we applied a forward voltage V_F ; The width w_0 is (potential barrier reduces) is reduced. When width reduces, electric field also reduces. Also, the potential reduces by the value of V_F . Therefore, potential will be equal to $V_0 - V_F$.

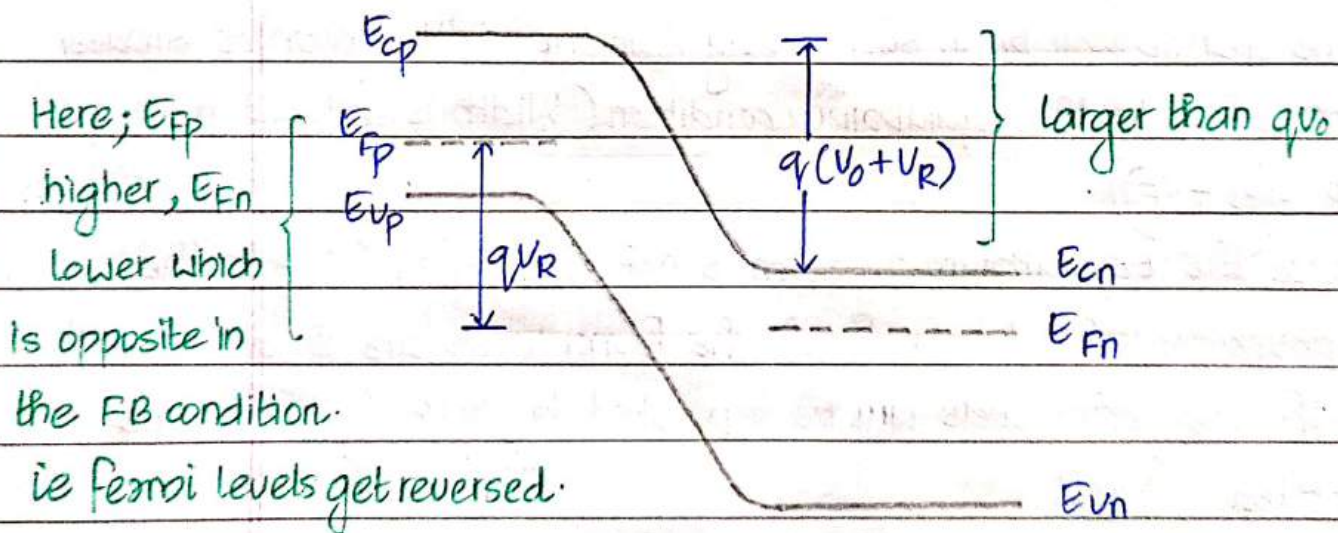
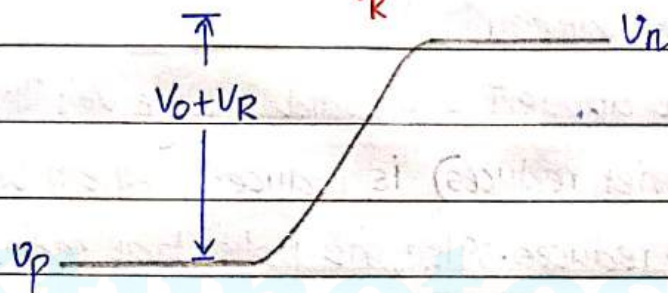
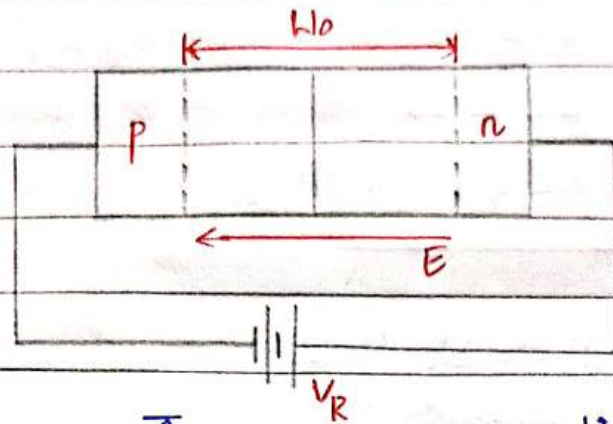
The change will be in such a way that the width becomes smaller compared to the equilibrium condition (Width is reduced from qV_0 to $q(V_0 - V_F)$).

→ Since the equilibrium condition is not maintained (due to the application of external field), the Fermi levels are separated and both the Fermi levels will be separated by qV_F (q times the applied voltage).

→ The applied voltage is +ve for F.B and the net voltage will be $V_0 - V_a$ i.e. $V_0 - V_F$ (a smaller value compared to V_0).

Note → Since the barrier is reduced, diffusion will be more, drift reduced. (drift current is less).

(iii) When reverse biased ($V_a = -V_R$)



→ The net potential (ie $V_0 + V_R$) will be larger than the eqm condition. Here -ve voltage is applied, ie $V_0 - (-V_R) = \underline{V_0 + V_R}$.

Particle flow	Current flow	
(1) $\rightarrow$	$\rightarrow$	(1) Hole diffusion
(2) $\leftarrow$	$\leftarrow$	(2) Hole drift
(3) $\leftarrow$	$\rightarrow$	(3) Electron diffusion
(4) $\dashrightarrow$	$\dashleftarrow$	(4) Electron drift

$\rightarrow$ Here, diffusion is shown by smaller arrows when compared to drift $\Rightarrow$ diffusion is less since the barrier is very high.

Diffusion by the majority carriers will be almost nil. Drift current is more compared to diffusion even though it is a small value, due to the movt of minority carriers.

(Current due to minority carriers (reverse current) is less a small value).

DISTRIBUTION OF CARRIER CONCENTRATION, POTENTIAL, ELECTRIC FIELD AND CHARGE DENSITY

Let, P_{p0} - Equilibrium hole concentration on the p-side.

n_{p0} - Equilibrium e^- concentration on the p-side.

n_{n0} - Equilibrium e^- concentration on the n-side.

p_{n0} - Equilibrium hole concentration on the n-side.

W_0 - Equilibrium depletion layer width.

x_{n0} - Equilibrium depletion layer width towards n-side.

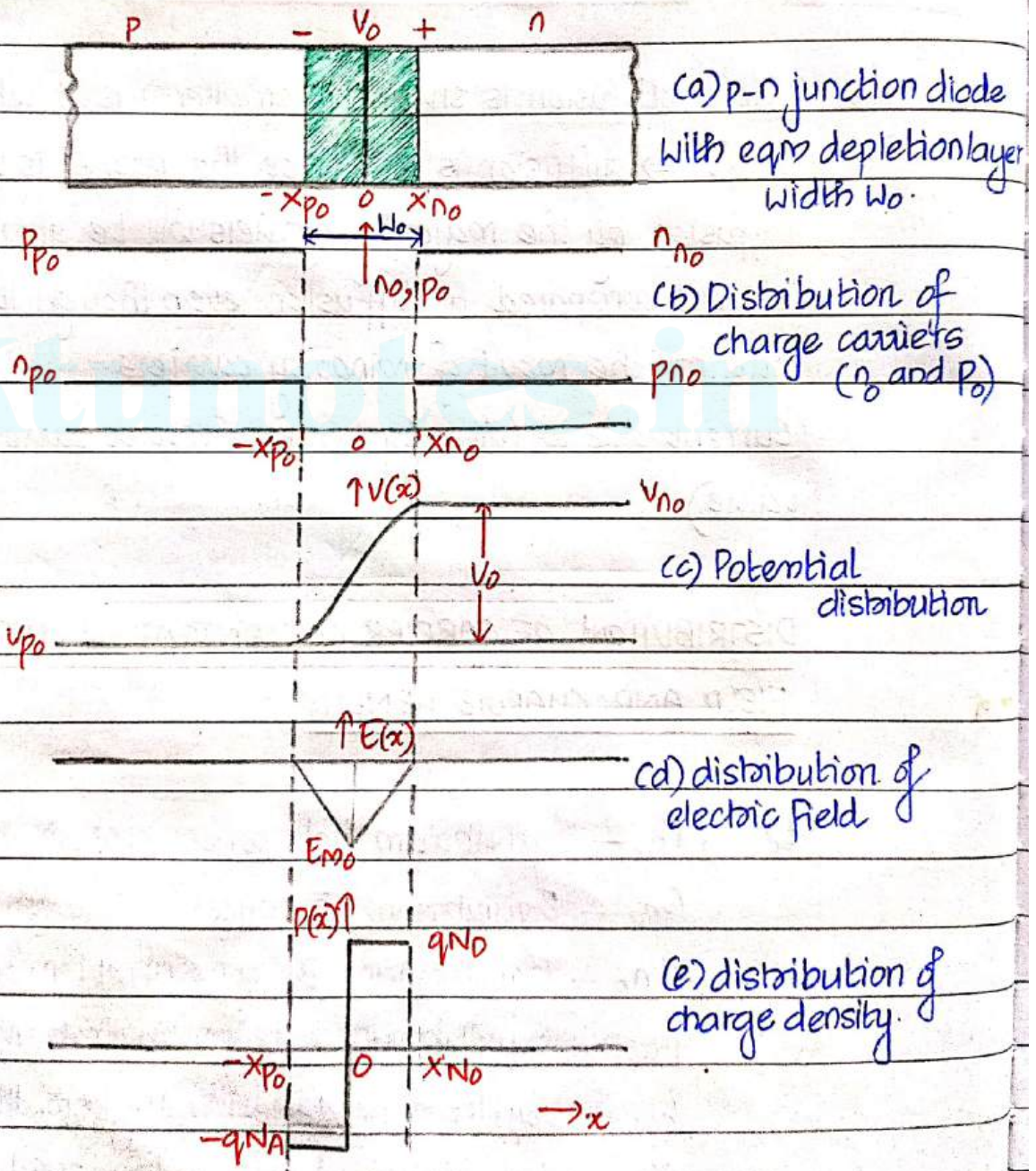
x_{p0} - Equilibrium depletion layer width towards p-side.

V_{n0} - Equilibrium potential at neutral n-side.

V_{p0} - Equilibrium potential at neutral p-side.

→ '0' present in all above terms indicates the eqbm value.

→ The term next to 0 (ie to the left) indicates the side (p or n side)



→ (e) Distribution of charge density - Positive charges are present on the n-side and -ve charges on the p-side due to the immobilized ions. If we assume that both are doped in equal amounts, we can say that charge distribution will be of symmetric form. However, charges will be equal on both sides irrespective of the doping on the n/p side; then only the charge neutrality can be maintained and proper width can be formed.

Hence, charge density will be equal on both side regardless of doping

Charge density = charge per unit volume = charge of ionized impurity * doping concentration.

$$\therefore \text{charge density on p-side} = -qN_A$$

$$\text{charge density on n-side} = +qN_D$$

The total charge on both sides are equal so that net charge is zero.

$$|Q_{Dp}| = |Q_{Dn}|$$

$$|Q_{Dp}| = \text{Total negative charge on the pside of depletion layer.}$$

$$|Q_{Dn}| = \text{Total +ve charge on the n side of depletion layer.}$$

$$|Q_{Dp}| = |\text{charge density}| \times \text{volume of depletion layer on p-side.}$$
$$= qN_A \times x_p \times A$$

Similarly, total charge on the n-side, $|Q_{Dn}| = qN_D X_{no} A$

$$\therefore qN_A X_{po} A = qN_D X_{no} A$$

$$\Rightarrow N_A X_{po} = N_D X_{no}$$

$$\frac{N_A}{N_D} = \frac{X_{no}}{X_{po}} \quad \text{--- (1)}$$

We also know that,

$$W_o = X_{no} + X_{po}$$

$$= \frac{N_A X_{po}}{N_D} + X_{po} \quad \text{by eqn. (1)}$$

$$= \left(\frac{N_A}{N_D} + 1 \right) X_{po}$$

$$= \left(\frac{N_A + N_D}{N_D} \right) X_{po}$$

$$\Rightarrow X_{po} = W_o \left[\frac{N_D}{N_A + N_D} \right] \quad \text{--- (A)}$$

$$\text{Similarly, } X_{no} = W_o \left[\frac{N_A}{N_A + N_D} \right] \quad \text{--- (B)}$$

From eqns (A) and (B), it can be inferred that X_{po} and X_{no} change w.r.t doping

Foreg, ~~we~~ consider that the acceptor ions are doped more, so that we will obtain a higher value of N_A . So in the eqn $X_{po} = W_o \left[\frac{N_D}{N_A + N_D} \right]$, denominator will be a higher value

and X_{po} gets reduced. In other words, if the p side doping (N_A)

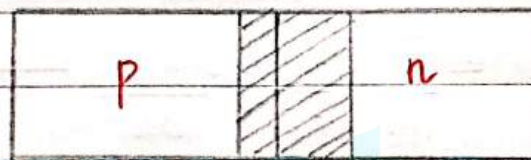
Is increased, width of the depletion layer on p-side gets reduced.



Increase in N_A results in a higher value of x_{n0} compared to x_{p0} .

Hence, the depletion layer width towards n-side will be higher compared to p-side if we increase the doping concn of p-type material.

For heavily doped p-material and lightly doped n-material.



$-x_{p0}$

qN_D

x_{n0}

distribution of charge density

Electric field.

E_{m0}

Remember that charges will be equal even though thickness is different.

THE BUILT IN POTENTIAL (V_0)

Under thermal equilibrium,

(total current = 0)

$$J_p = 0 \text{ and } J_n = 0$$

$$\Rightarrow J_p = q p_0(x) \mu_p E(x) - q D_p \frac{dp_0(x)}{dx} = 0$$

From above equation,

$$q p_0(x) \mu_p E(x) = q D_p \frac{dp_0(x)}{dx}$$

$$E(x) = \frac{D_p}{\mu_p} \times \frac{1}{p_0(x)} \frac{dp_0(x)}{dx} \text{ ——— ①}$$

$$\text{But } \frac{D_p}{\mu_p} = \frac{kT}{q} \text{ and } E(x) = -\frac{dV}{dx}$$

$\therefore$ Eqn ① reduces to

$$-\frac{dV}{dx} = \frac{kT}{q} \times \frac{1}{p_0(x)} \times \frac{dp_0(x)}{dx}$$

$$-\int dV = \int \frac{kT}{q} \cdot \frac{1}{p_0(x)} dp_0(x)$$

LHS - Integrate wrt potential

RHS - Integrate wrt hole concentration

$$-\int_{V_{p0}}^{V_{n0}} dV = \frac{kT}{q} \int_{p_{p0}}^{p_{n0}} \frac{dp_0(x)}{p_0(x)}$$

$$-[V_{no} - V_{po}] = \frac{kT}{q} \ln \left[\frac{p_o(x)}{p_{po}} \right]^{p_{no}}$$

$$= \frac{kT}{q} [\ln p_{no} - \ln p_{po}]$$

$$V_{no} - V_{po} = \frac{kT}{q} [\ln p_{po} - \ln p_{no}]$$

$$V_o = V_{no} - V_{po} = \frac{kT}{q} \ln \frac{p_{po}}{p_{no}} \quad \text{--- (2)}$$

But $p_{po} = N_A$ (Eqbm majority carrier conc'n on p side)

$n_{no} = N_D$ (Eqbm majority carrier conc'n on n-side)

$p_{no} = \frac{n_i^2}{N_D}$ (Eqbm minority carrier conc'n on n side)

$$\therefore \text{The above eqn becomes, } V_o = \frac{kT}{q} \ln \frac{N_A N_D}{n_i^2} \quad \text{--- (3)}$$

$\therefore$ Eqn (2) can also be written as

$$\frac{p_{po}}{p_{no}} = e^{qV_o/kT}$$

If we start with $T_n = 0$; it can be shown that

$$\frac{n_{no}}{n_{po}} = e^{qV_o/kT}$$

- Effect of T and doping conc'n on eqn (3) i.e. cut-in voltage:
- $\rightarrow$ From eqn 3, it can be inferred that Increase in doping conc'n (say, N_A or N_D) results in an Increase in contact potential, V_o .

→ Also, at a constant T and doping conc'n, let's assume that the band gap is increased. So, when we increase the band gap, when we consider a band gap of a particular slc of increased quantity, n_i^2 value becomes less. If n_i^2 is less, the cut in voltage will be more.

• $\text{band gap} \uparrow \Rightarrow n_i^2 \downarrow \Rightarrow V_0 \uparrow$

Eg: Cut in voltage of Si > Ge since there is a difference in band gap.

→ Let's assume that temperature T is varied. Although $V_0 \propto T$, temperature affects the n_i value. So, if T is increased, n_i also gets increased and this increase is large compared to that of T . (very large compared to normal T).



Since n_i increases abundantly, V_0 decreases on increase in T even though V_0 and T are directly proportional.

ELECTRIC FIELD

Start with poisson's eqn

$$\frac{dE(x)}{dx} = \frac{\rho}{\epsilon} \quad (\text{relates gradient in electric field to charge density } \rho)$$

$\epsilon \rightarrow$ relative permittivity

$$= \frac{q}{\epsilon} (N_D^+ - N_A^- + p - n)$$

By depletion approximation, $p = n = 0$

$\Rightarrow$

$$\begin{aligned} \frac{dE(x)}{dx} &= \frac{q}{\epsilon} N_D ; 0 < x < x_{n0} \\ \frac{dE(x)}{dx} &= -\frac{q}{\epsilon} N_A ; -x_{p0} < x < 0 \end{aligned} \quad \text{--- ①}$$

For $-x_{p0} < x < 0$ (p side)

($N_A^+ = N_A \rightarrow$ completely depleted)

$$\rho = -qN_A$$
$$\frac{dE(x)}{dx} = -\frac{qN_A}{\epsilon}$$

$$\int_0^{E_{m0}} dE(x) = \int_{-x_{p0}}^0 -\frac{q}{\epsilon} N_A dx$$

$$(E_{m0} - 0) = -\frac{q}{\epsilon} N_A x \Big|_{-x_{p0}}^0$$

$$\Rightarrow E_{m0} = -\frac{q}{\epsilon} N_A [0 - (-x_{p0})]$$

$$E_{m0} = -\frac{q}{\epsilon} N_A x_{p0} \quad \left[\text{max. value of electric field} \right]$$

On n-side i.e. $0 < x < x_{n0}$

$$\int_{E_{m0}}^0 dE(x) = \int_0^{x_{n0}} \frac{q}{\epsilon} N_D dx$$

$$(0 - E_{m0}) = \frac{q}{\epsilon} N_D [x] \Big|_0^{x_{n0}}$$

$$-E_{m0} = \frac{q}{\epsilon} N_D x_{n0}$$

$$\Rightarrow E_{m0} = -\frac{q}{\epsilon} N_D x_{n0}$$

CONTACT POTENTIAL WRT W_0

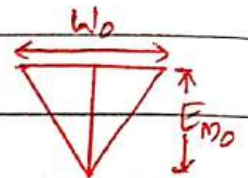
$$E(x) = -\frac{dV(x)}{dx} = -\int_{V_{p0}}^{V_{n0}} dV(x) = \int_{-x_{p0}}^{x_{n0}} E(x) dx$$

$$-(V_{n0} - V_{p0}) = \int_{-x_{p0}}^{x_{n0}} E(x) dx$$

$$-V_0 = \int_{-x_{p0}}^{x_{n0}} E(x) dx$$

$$\therefore -V_0 = \frac{1}{2} E_{m0} W_0$$

($\frac{1}{2} \times \text{base} \times \text{height}$)



$$\Rightarrow V_0 = -\frac{1}{2} E_{m0} W_0$$

Width of depletion layer:

$$V_0 = -\frac{1}{2} E_{m0} W_0$$

$$E_{m0} = -\frac{q}{\epsilon} N_A x_{p0}$$

$$\therefore V_0 = -\frac{1}{2} \times -\frac{q}{\epsilon} N_A x_{p0} W_0$$

$$= \frac{q}{2\epsilon} N_A x_{p0} W_0$$

$$x_{p0} = \frac{W_0 \cdot N_D}{N_A + N_D}$$

$$\therefore V_0 = \frac{q}{2\epsilon} \frac{N_A N_D}{N_A + N_D} \cdot W_0^2$$

$$\begin{aligned}
 W_0^2 &= \frac{2\epsilon V_0}{q} \left(\frac{N_A + N_D}{N_A N_D} \right) \\
 &= \frac{2\epsilon V_0}{q} \left(\frac{N_A}{N_A N_D} + \frac{N_D}{N_A N_D} \right) \\
 &= \frac{2\epsilon V_0}{q} \left(\frac{1}{N_D} + \frac{1}{N_A} \right) \\
 \Rightarrow W_0 &= \sqrt{\frac{2\epsilon V_0}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right)} \quad \text{--- (c)}
 \end{aligned}$$

There are several variations of eqn c,

For eg $V_0 = \frac{kT}{q} \ln \frac{N_A N_D}{n_i^2}$

$$\therefore W_0 = \sqrt{\frac{2\epsilon kT}{q^2} \left(\ln \frac{N_A N_D}{n_i^2} \right) \left[\frac{1}{N_A} + \frac{1}{N_D} \right]}$$

We can also calculate the penetration of the transition region into the n & p materials.

We know that $X_{p0} = W_0 \cdot \frac{N_D}{N_A + N_D} = \frac{W_0}{1 + \frac{N_A}{N_D}} = \sqrt{\frac{2\epsilon V_0}{q} \left[\frac{N_D}{N_A(N_A + N_D)} \right]}$

$$X_{n0} = W_0 \frac{N_A}{N_A + N_D} = \frac{W_0}{1 + \frac{N_D}{N_A}} = \sqrt{\frac{2\epsilon V_0}{q} \left[\frac{N_A}{N_D(N_A + N_D)} \right]}$$

The above eqns show the relationship of width with the doping conc'n and applied voltage.

Application of bias voltage on junction parameter

→ If we apply an external voltage V_a , the V_0 value will change from $V_0 \rightarrow V_0 - V_a$. The applied voltage V_a is +ve in the case of forward bias and -ve for reverse bias.

$$V_0 - V_F : \text{FB} ; V_0 + V_R : \text{RB}$$

$$\rightarrow W_0 \rightarrow W = \sqrt{\frac{2\epsilon(V_0 - V_a)}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right)}$$

→ On the application of +ve voltage / F-B voltage, the effective voltage will be reduced ($V_0 - V_a$) and hence, width will be reduced. Similarly, on the application of R-B voltage, width $\uparrow$.

$$\rightarrow X_{p0} \rightarrow X_p = \frac{W \cdot N_D}{N_A + N_D}$$

$$X_{n0} \rightarrow X_n = \frac{W \cdot N_A}{N_A + N_D}$$

$$\rightarrow E_{m0} \rightarrow E_m = -\frac{q}{\epsilon} N_A X_p = -\frac{q}{\epsilon} N_D X_n$$

Ideal diode equation

(Here, we express everything in terms of minority carriers i.e. the minority carrier conc'n on p-side and n-side).

The equilibrium ratio of hole conc'n on each side is given by

$$\frac{P_{p0}}{P_{n0}} = e^{qV_0/kT} \quad \text{--- ①}$$

With bias, $V_0 \rightarrow V_0 - V_a$

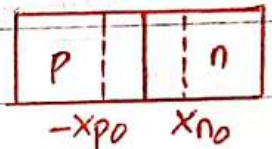
$$P_{p0} \rightarrow P_p(-x_{p0})$$

$$P_{n0} \rightarrow P_n(x_{n0})$$

→ represents the edge of depletion layer on p.

∴ ① can be written as

$$\frac{P_p(-x_{p0})}{P_n(x_{n0})} = e^{q(V_0 - V_a)/kT} \quad \text{--- ②}$$



For low level injection, we can neglect any change in the majority carrier conc'n. → case where injected excess carrier conc'n is

majority carrier concentration on p-side. ∴ $P_p(-x_{p0}) = P_{p0}$ very low compared to eqbm conc'n.

①
②

$$\frac{P_{p0}}{P_{n0}}$$

$$= \frac{e^{qV_0/kT}}{\frac{P_p(-x_{p0})}{P_n(x_{n0})}}$$

In our case, it exists under forward biased condition.

So, low level injection is a term used

$$\frac{P_n(x_{n0})}{P_{n0}} = e^{qV_0/kT} \cdot e^{-q(V_0 - V_a)/kT}$$

for a diode under F.B condition.

minority carrier conc'n

on n-side ∴ $P_n(x_{n0}) = P_{n0} e^{qV_a/kT} \quad \text{--- ③}$

∴ If V_a is +ve (F.B voltage), $e^{qV_a/kT}$ increases, and hence the minority carrier conc'n increases, compared to the eqbm state. A similar

case is observed when we consider $n_p(x_{p0})$, where the minority carrier concentration at p-side increases.

→ So, under FB condition, minority carrier conc'n ↑ and under R-B condition, minority carrier conc'n is low compared to the eqbm condition.

Here, the injected excess minority carrier conc'n is represented as:

$\Delta p_n \rightarrow$ Excess minority hole concentration @ x_{n0}

$\Delta n_p \rightarrow$ Excess minority $\bar{e}$ concentration @ x_{p0}

The total hole conc'n at x_{n0} is given as

$$p_n(x_{n0}) = p_{n0} + \Delta p_n$$

$$\therefore \Delta p_n = p_n(x_{n0}) - p_{n0} = p_{n0} e^{qV_a/kT} - p_{n0}$$

$$\Delta p_n = p_{n0} (e^{qV_a/kT} - 1) \quad \text{--- (4)}$$

$$\text{Similarly, } \Delta n_p = n_{p0} (e^{qV_a/kT} - 1) \quad \text{--- (5)}$$

Note:- (Why we are expressing the above conditions in terms of minority carrier concentration?)

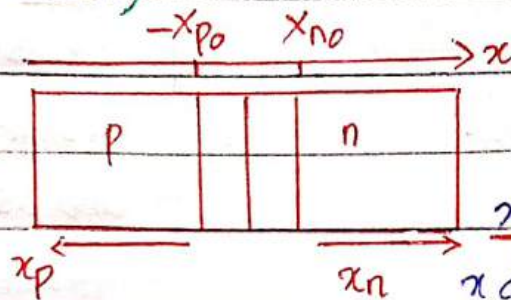
Even though the majority carriers from p side and n side are crossing the junction; on reaching the opposite sides, they become minority carriers.



For eg; holes are majority carriers on

the p side, but on reaching the opposite side, they become minority carriers. This is the reason why express everything in terms of

minority carriers)



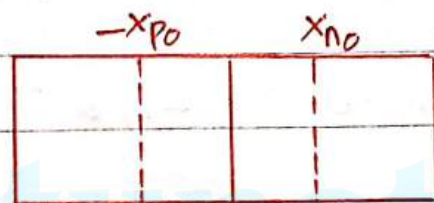
x_n is: distances measured in the x -direction in the n -material from x_{n0} will be designated as " x_n ".

x_p : distances in the p -material in x direction with $-x_{p0}$ as origin.

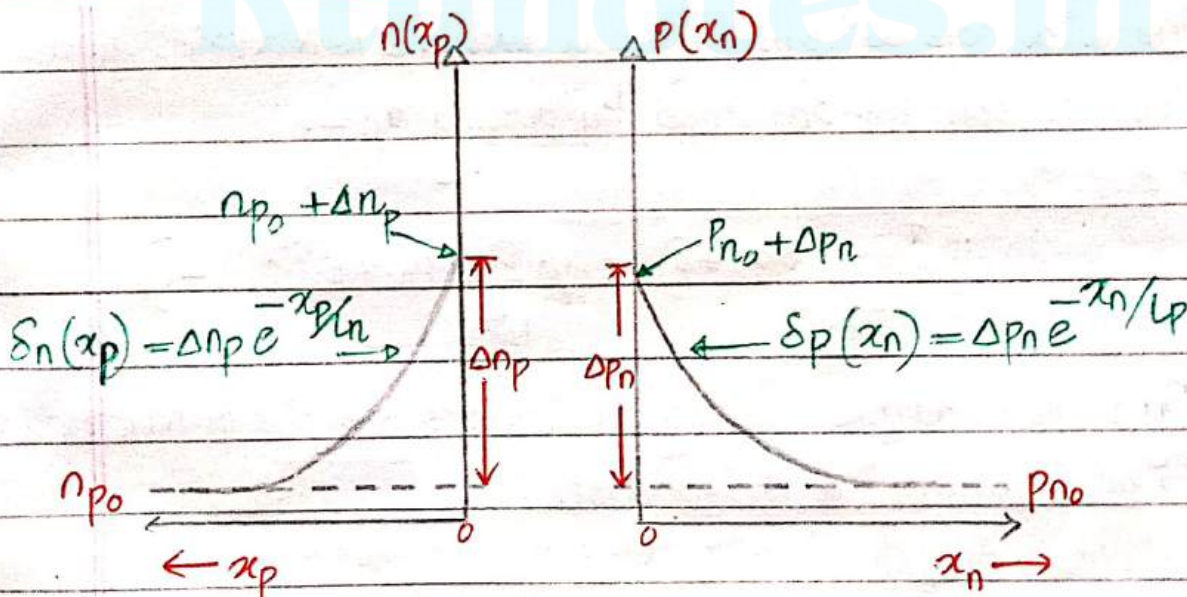
Excess carrier distribution on long n side and p side is as follows.

$$\delta p(x_n) = \Delta p_n e^{-x_n/L_p} \quad (\text{minority carriers are holes on } n\text{-side})$$

$$\delta n(x_p) = \Delta n_p e^{-x_p/L_n} \quad (\text{minority carriers are } \bar{e}\text{s on } p\text{-side}).$$



From the above eqns, We can observe that the injected excess carrier $\downarrow$ exponentially.



(Fig) Minority carrier distribution on both p and n sides across a FB $p-n$ jn.

$\rightarrow$ The minority carrier distribution currents on both sides:

The hole diffusion current on n side is

$$I_{p \text{ diff}}(x_n) = -qA D_p \frac{d \delta p(x_n)}{dx_n}$$

$$= -qA D_p \frac{d}{dx_n} (\Delta p_n e^{-x_n/L_p})$$

$$= qA D_p \frac{\Delta p_n}{L_p} e^{-x_n/L_p}$$

Substitute for Δp_n ,

$$\therefore I_{p \text{ diff}}(x_n) = qA D_p \frac{p_{n0}}{L_p} (e^{qV_a/kT}$$

$$- 1) e^{-x_n/L_p}$$

A: cross sectional area of jn.

$$\rightarrow I_{p \text{ diff}}(x_n) = qA D_p \frac{p_{n0}}{L_p} (e^{qV_a/kT} - 1) e^{-x_n/L_p}$$

The total hole current injected into the n-material at the junction can be obtained by putting $x_n = 0$

$$\therefore I_{p \text{ diff}}(x_n = 0) = qA \frac{D_p}{L_p} p_{n0} (e^{qV_a/kT} - 1)$$

By similar analysis, the injection of $\bar{e}$'s into the p-material leads to an $\bar{e}$ current at the junction.

$$I_{n \text{ diff}}(x_p = 0) = -qA \frac{D_n}{L_n} n_{p0} (e^{qV_a/kT} - 1)$$

$$\Rightarrow \bar{I}_{n \text{ diff}}(x_p = 0) = qA \frac{D_n}{L_n} n_{p0} (e^{qV_a/kT} - 1)$$

The minus sign indicates that the actual current flows in

► Note: Δn_p and Δp_n are the excess $\bar{e}$ s and holes on the p-side and n-side respectively and as time varies, it decreases as per the eqns of $\delta p(x_n)$ and $\delta n(x_p)$.

opposite direction.

The total diode current $I = I_{p \text{ diff}}(x_n=0) + (-I_{n \text{ diff}}(x_p=0))$

$$= qA \left[\frac{D_p}{L_p} P_{n0} + \frac{D_n}{L_n} n_{p0} \right] \left[e^{qV_a/kT} - 1 \right]$$

$$\therefore I = I_0 (e^{qV_a/kT} - 1) \quad \text{--- (6)}$$

The above equation is known as **diode equation**.

I_0 - Reverse saturation current.

- If V_a is a +ve voltage / F.B voltage, current $\uparrow$ exponentially.
- If V_a is -ve / R.B condition, $V_a = -V_R$

$|V_R| \gg \frac{kT}{q}$ then,

"Total current is just the reverse saturation current."

$$I = -I_0$$

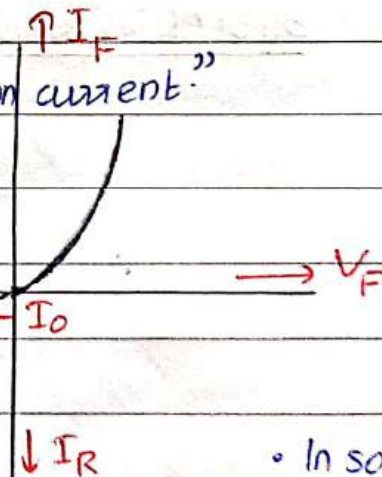
Minority carrier current exists and that leakage current is shown by the value I_0 .

$$I = I_0 (e^{-\infty} - 1)$$

$$= I_0 (-1)$$

$$= -I_0$$

==



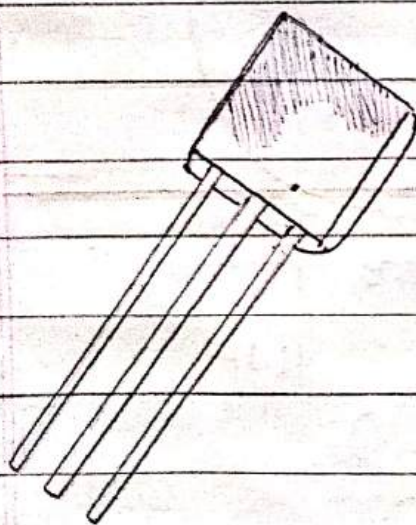
• In some text books, I_0 is also referred as I_s .

→ If we increase the reverse voltage, the junction breakdowns and all the carriers will cross the junction; the current flow will be very large.

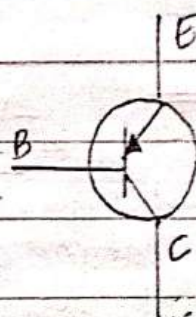
Note:- With $V_a > 0$, the diode is forward biased and current $\uparrow$ exponentially with $\uparrow$ in forward voltage. The change in forward current is large for a small change in the forward voltage. $\therefore$ The resistance of diode under forward-biased condition is very small.

→ The reverse saturation current I_0 is very small current and is of the order of several μAmp for Ge diode and nano amperes for a Si diode.

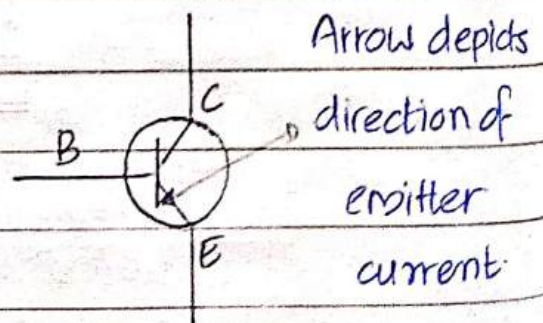
BIPOLAR JUNCTION TRANSISTORS



Typical bipolar
junction transistor



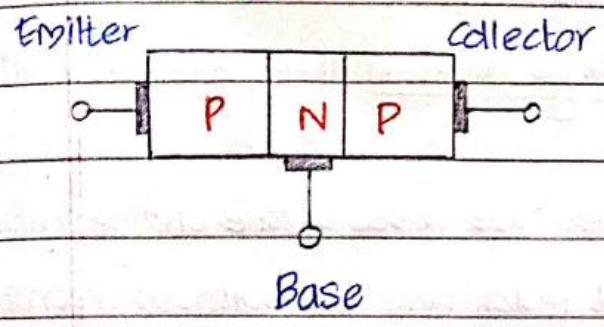
PNP BJT



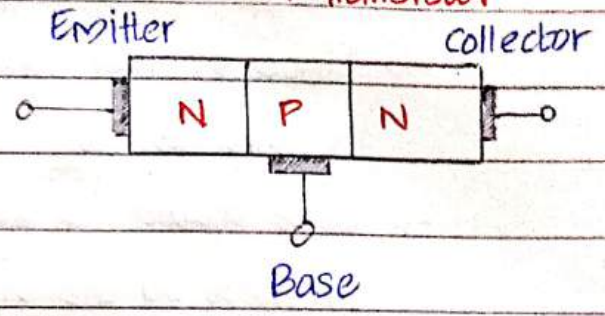
NPN BJT

→ Discovered by John Bardeen, William Shockley and Walter Brattain at Bell Labs in 1948.

PNP Transistor

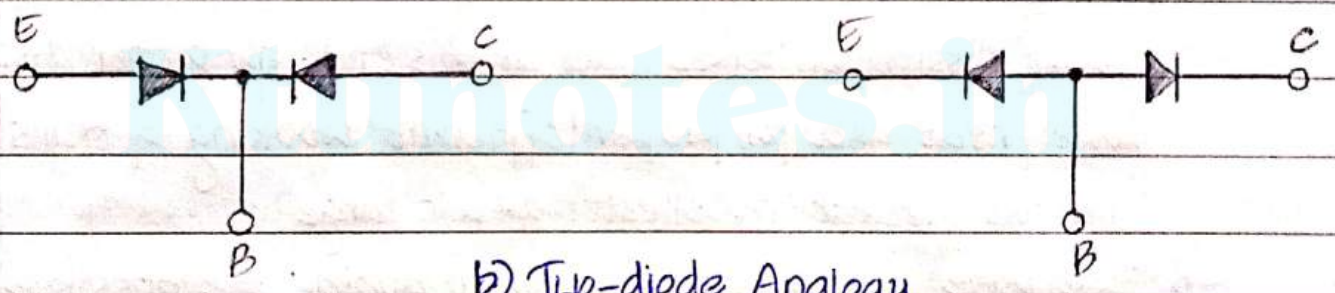


NPN Transistor

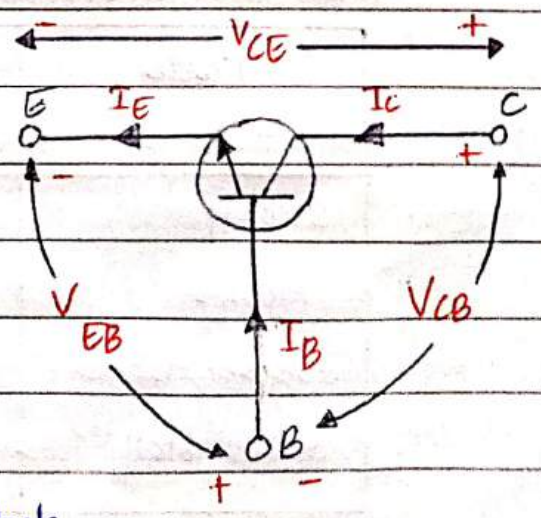
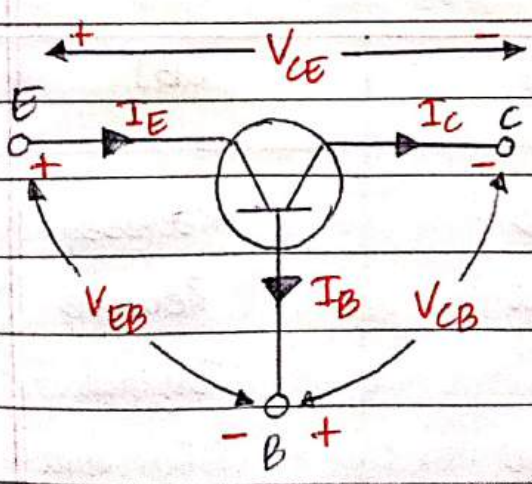


a) Physical Construction

(Two diodes connected back to back).



b) Two-diode Analogy



c) Circuit symbols

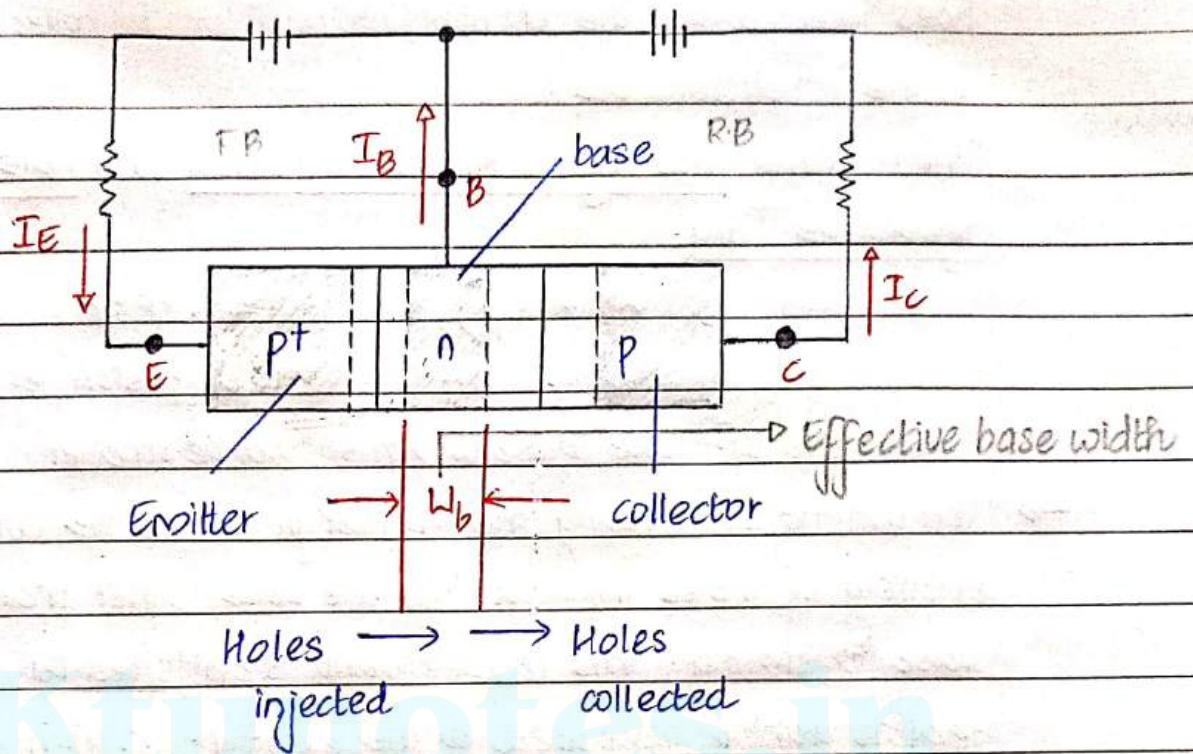
- Emitter emits the charge carriers to collector and collector collects the charge carriers; Base is the region through which the emitted charge carriers flow towards the collector.
- Since emitter needs to emit large no. of charge carriers, it should be heavily doped.
- Base should be lightly doped because most of the charge carriers emitted by the emitter should reach collector without recombination. Hence, size of base and its doping should be less (very lightly doped).
- collector have a doping level in b/w emitter and base.
Collector area is very large compared to base and emitter bcz power dissipation takes place at this end. For proper dissipation of power/heat and for proper/efficient working of transistor without breakdown, collector area is large.
- The base area has a is small to avoid the possible recombination of the emitted charge carriers. from

MODES OF OPERATION

Mode	EBT	CBT
Cut off	Reverse	Reverse
Active	Forward	Reverse
Reverse Active	Reverse	Forward
Saturation	Forward	Forward.

Normally, transistors works in the active region.

FUNDAMENTALS OF OPERATION



→ + symbol in p^+ indicates that it is heavily doped. Since there is diff in the doping conc'n's, we can say that the depletion region across these junctions also varies.

(Smaller depletion region on the left side due to heavy doping)

→ The injection of holes from $p^+ - n$ into the centre n region supplies the minority carrier holes to participate in the reverse current through n-p.

{ In order to restrict holes from recombining with e^- s in the n region, n region should be narrow compared to hole diffusion length }

→ The above fig is in CB configuration.

- To have a good P-N-P transistor, we must prefer that almost all holes injected by the emitter into collector be collected by collector.
 $\therefore$ The conditions are:

The n-type base region should be narrow and hole life time τ_p should be long.

$$(W_b \ll L_p)$$

$$L_p = \sqrt{D_p \tau_p}$$

L_p is diffusion length for holes.

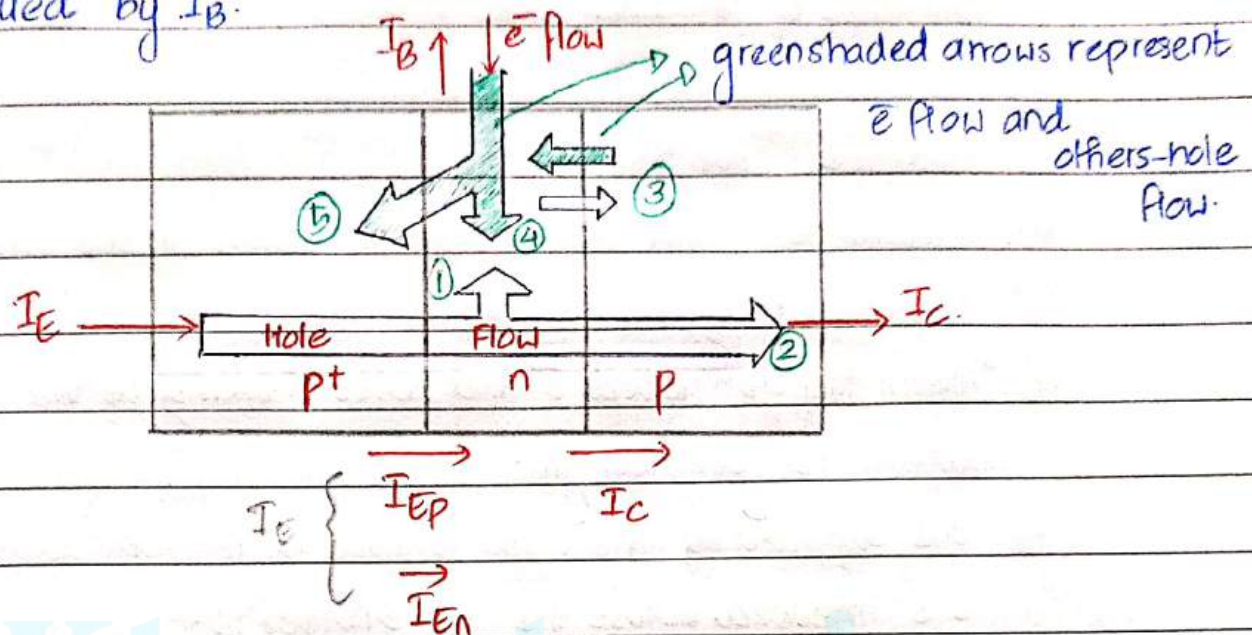
Thus the effective base width should be smaller.

- The current I_E crossing the emitter-jn should be composed almost entirely of holes injected into the base, rather than $\bar{e}$ crossing from base to emitter. This requirement is satisfied by doping the base region lightly compared to the emitter. $\therefore$ p⁺-n emitter-jn results.
- The current I_E flows into the emitter of a properly biased p-n-p transistor and that I_C flows out at the collector.
- In a good transistor, the base current I_B will be very small since I_E is essentially hole current and collected hole current I_C is almost equal to I_E .

CURRENT COMPONENTS

- There must be some base current due to the requirement of electron flow into n-type base region. 
- We can account for I_B physically by following mechanisms:
- (a) There must be some recombination of injected holes with $\bar{e}$'s in base even with $W_b \ll L_p$. The $\bar{e}$'s lost due to recombination must be resupplied through the base contact.

(b) Some $\bar{e}$'s will be injected from n to p in the F-B emitter even if emitter is heavily doped compared to base. These $\bar{e}$'s must also be supplied by I_B .



- (1) Injected holes lost to recombination in the base.
- (2) Holes reaching the R-B collector junction.
- (3) Thermally generated $\bar{e}$ s and holes making up the reverse saturation current of the collector junction.
- (4) Electrons supplied by the base contact for recombination with holes.
- (5) $\bar{e}$ s injected across the forward-biased emitter junction.



→ The dominant sources are of base current:

- (1) Recombination in base
- (2) Injection into emitter region.

→ In an n-p-n transistor, the 3 current directions are reversed.
The mechanism of npn can be understood by simply reversing the roles of $\bar{e}$'s and holes in p-n-p.

From the above figure:

- (1) Injected holes lost due to recombination in the base.
- (2) Holes reaching the R.B collector jn.
- (3) Thermally generated $\bar{e}$ and holes making up the reverse saturation current of collector jn.
- (4) $\bar{e}$'s supplied by the base contact for recombination with holes.
- (5) $\bar{e}$'s injected across the FB emitter jn.

AMPLIFICATION

$$i_c = B i_{Ep} \quad B = \text{Base transportation factor.}$$

The emitter injection efficiency γ ,

$$\gamma = \frac{i_{Ep}}{i_{En} + i_{Ep}}$$

The relation b/w the collector and emitter currents is

$$\frac{i_c}{i_E} = \frac{B i_{Ep}}{i_{En} + i_{Ep}} = B \gamma = \alpha$$

The product $B\gamma$ is defined as the factor α , called the current transfer ratio, which represents the emitter-to-collector

current amplification. There is no real amplification between these currents, since α is smaller than unity. On the other hand, the relation b/w i_c and i_b is more promising for amplification.

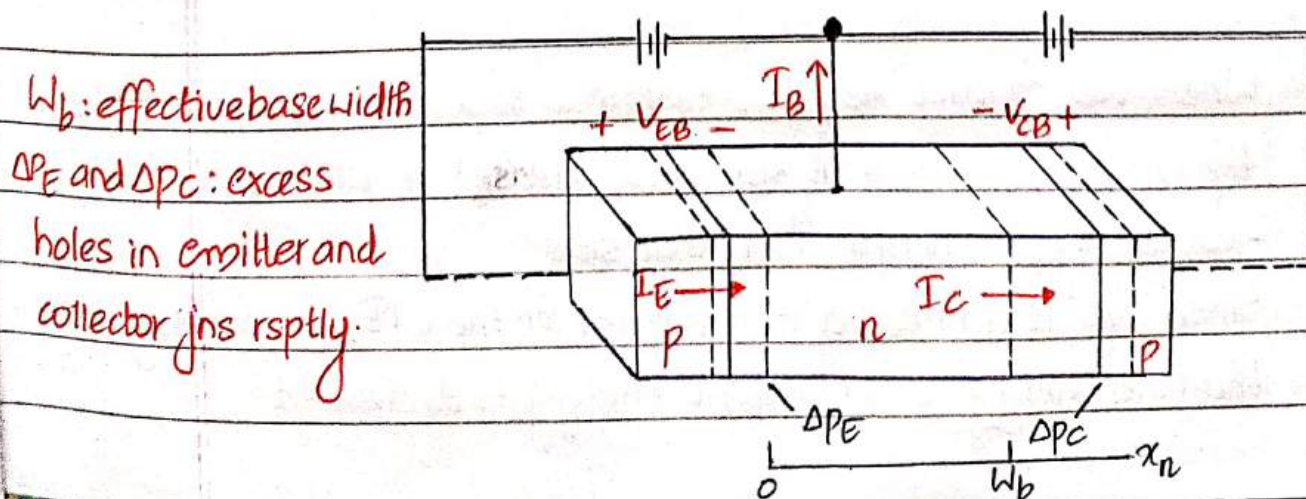
The total base current, $I_B = i_{En} + (1-\beta)i_{Ep}$

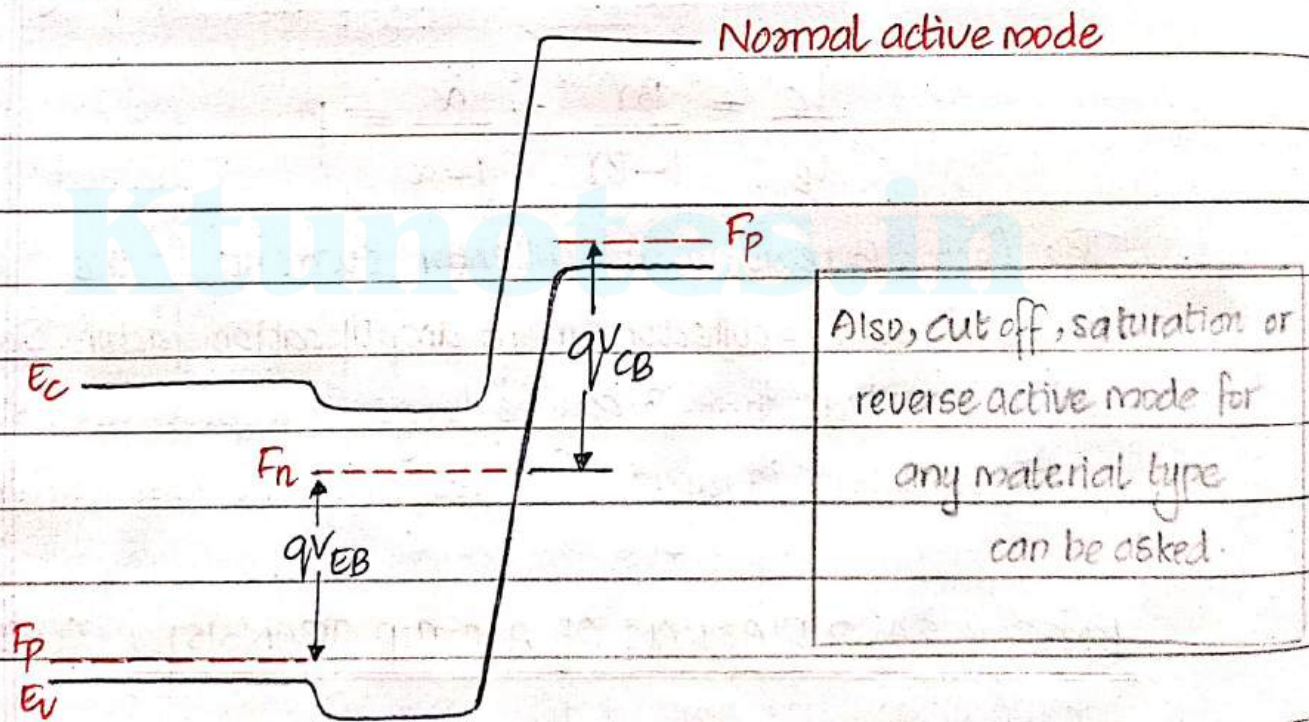
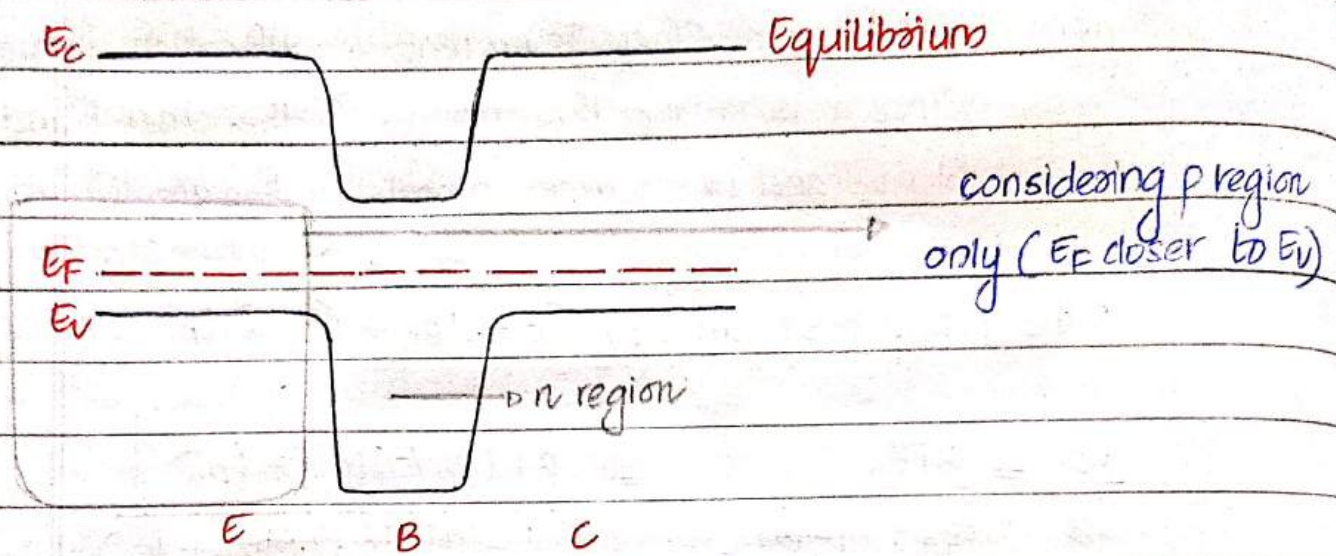
$$\frac{i_c}{i_b} = \frac{\beta i_{Ep}}{i_{En} + (1-\beta)i_{Ep}} = \frac{\beta [i_{Ep}/(i_{En} + i_{Ep})]}{1 - \beta [i_{Ep}/(i_{En} + i_{Ep})]}$$

$$\frac{i_c}{i_b} = \frac{\beta \gamma}{1 - \beta \gamma} = \frac{\alpha}{1 - \alpha} \equiv \beta$$

The factor β relating the collector current to the base current is the base-to-collector current amplification factor. Since α is near unity, it is clear that β can be large for a good transistor, and the collector current is large compared with the base current.

ENERGY BAND DIAGRAM OF A P-N-P TRANSISTOR (under the eqbm condition and the normal active mode).





- When we connect negative terminal to a side, we can say that the energy level of that side gets increased i.e. when there is a -ve supply, $\bar{e}$ energy $\uparrow$ on that side.
- Since +ve is connected to the p-side (on the left); we can say that the hole energy is increased / $\bar{e}$ energy is decreased.

→ So there will be a shift in the energy band. i.e;

Eqbm condition: There was a large variation in the conduction band

When biasing is given, the Fermi level changes to quasi Fermi level and there will be a shift in this level; [$\bar{E}$ energy $\uparrow$ in the n region]
* there will be a shift upwards

the terminal is connected to p side so there is a decrease in $\bar{E}$ energy. So, that particular region shifts little downwards.

→ n energy levels shift $\uparrow$ and p energy levels shift $\downarrow$. And if the condition of qV_{EB} (the diff b/w Fermi levels) is satisfied, this shift stops.

→ On the other side, the terminal is connected to n-side and -ve to p. So the p levels shift $\uparrow$ and n $\downarrow$ losing the $\bar{E}$ energy. Diff b/w F_n and F_p will be qV_{CB} .

→ n-p-n transistor: - A flipped version of the above condition.

MINORITY CARRIER DISTRIBUTION AND TERMINAL CURRENTS

We assume that the holes are injected into the base at the F.B

emitter and these holes diffuse to the collector junction.

→ In order to know the terminal currents:

- ▶ First step is to solve for excess hole distribution in the base
- ▶ Second step is to evaluate the emitter and collector currents (I_E and I_C) from gradient of hole distribution on each side of the base.

The base current can be found from current summation or from a charge control analysis of recombination in the base.

→ We can simplify the calculations by making several assumptions:

- ▶ Holes diffuse from emitter to collector; drift is negligible in base region.
- ▶ Emitter current is entirely made up of holes; emitter injection efficiency, $\gamma = 1$

$$\gamma = \frac{i_{Ep}}{i_{Ep} + i_{En}} \rightarrow \begin{array}{l} \text{the hole component of total emitter} \\ \text{current} \end{array}$$

→ The $\bar{e}$ component of total emitter current (due to $\bar{e}$ s emitted from B to E)

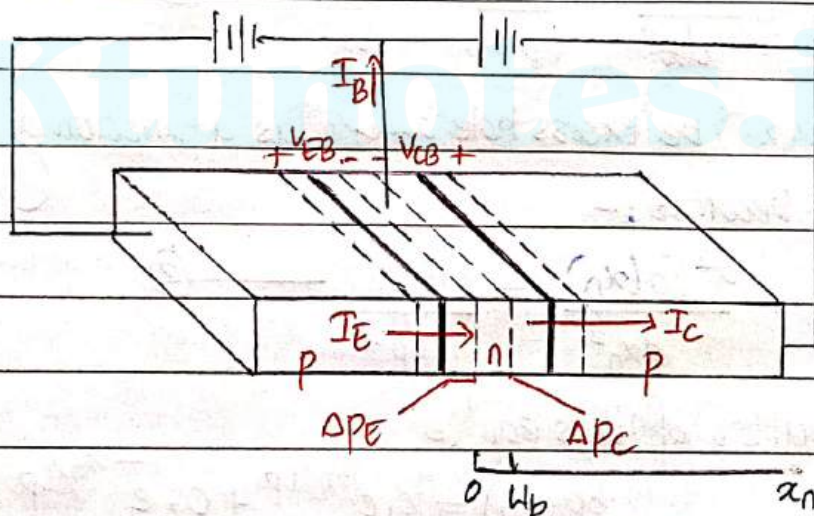
- ▶ The collector saturation current is negligible.
- ▶ Current flow in the base is essentially one dimensional from emitter to collector.
- ▶ All currents and voltages are steady state.

The active part of the base and the two junctions are of uniform cross-sectional area A .

SOLUTION OF DIFFUSION EQN IN THE BASE REGION

Neglecting the recombination in 2 depletion regions, hole current entering the base at the emitter junction is current I_E and leaving the base at the collector junction is current I_C .

- If we can solve for the distribution of excess holes in the base region, it is simple to evaluate the gradient of distribution at 2 ends of the base to find the currents.
- Following is the simplified p-n-p transistor geometry used in the calculations:-



- Here the base width W_b is b/w 2 depletion regions and uniform cross sectional area is A .
- The barrier at E-B jn is reduced by FB and that at C-B jn is ↑ by R-B.
- ΔP_E and ΔP_C → Excess hole conc'n at edge of emitter side and collector side respectively.

$$\Delta P_E = P_{n0} (e^{qV_{EB}/KT} - 1) \quad P_{n0} - \text{Eq pro hole conc'n}$$

$$\Delta P_C = P_{n0} (e^{qV_{CB}/KT} - 1)$$

→ If the E-B junction is strongly F.B ($V_{EB} \gg \frac{KT}{q}$) and the CB junction is strongly R.B ($V_{CB} \gg 0$) these excess hole conc'n simplified to

$$\Delta P_E \cong P_{n0} e^{qV_{EB}/KT} \quad \text{--- (a)}$$

$$\Delta P_C \cong -P_{n0} \quad \text{--- (b)}$$

We have assumed that the current and voltages are steady state

In steady state case, the diffusion eqn becomes,

$$\frac{d^2 \delta p}{dx^2} = \frac{\delta p}{D_p \tau_p} = \frac{\delta p}{L_p^2} \quad \text{--- (1)}$$

Let $\delta p(x_n)$ be excess hole conc'n as a function of distance.
Eqn (1) becomes :-

$$\frac{d^2 \delta p(x_n)}{dx_n^2} = \frac{\delta p(x_n)}{L_p^2} \quad \text{--- (2)}$$

The solution of this eqn is

$$\delta p(x_n) = C_1 e^{x_n/L_p} + C_2 e^{-x_n/L_p} \quad \text{--- (3)}$$

$L_p \rightarrow$ diffusion length of holes in base region.

In this case, the appropriate boundary conditions are

$$\delta p(x_n = 0) = C_1 + C_2 = \Delta P_E$$

$$\delta p(x_n = W_b) = C_1 e^{W_b/L_p} + C_2 e^{-W_b/L_p} = \Delta P_C$$

Solving the parameters C_1 and C_2

$$C_1 = \Delta P_E - C_2$$

$$\therefore (\Delta P_E - C_2) e^{W_b/L_p} + C_2 e^{-W_b/L_p} = \Delta P_C$$

$$\Delta P_E e^{W_b/L_p} - C_2 e^{W_b/L_p} + C_2 e^{-W_b/L_p} = \Delta P_C$$

$$\Delta P_E e^{W_b/L_p} - \Delta P_C = C_2 e^{W_b/L_p} - C_2 e^{-W_b/L_p}$$

$$\Rightarrow C_2 (e^{W_b/L_p} - e^{-W_b/L_p}) = \Delta P_E e^{W_b/L_p} - \Delta P_C$$

$$C_2 = \frac{\Delta P_E e^{W_b/L_p} - \Delta P_C}{e^{W_b/L_p} - e^{-W_b/L_p}}$$

$$C_1 = \Delta P_E - C_2$$

$$= \Delta P_E - \frac{(\Delta P_E e^{W_b/L_p} - \Delta P_C)}{e^{W_b/L_p} - e^{-W_b/L_p}}$$

$$= \frac{\Delta P_E e^{W_b/L_p} - \Delta P_E e^{-W_b/L_p} - \Delta P_E e^{W_b/L_p} + \Delta P_C}{e^{W_b/L_p} - e^{-W_b/L_p}}$$

$$\Rightarrow C_1 = \frac{\Delta P_C - \Delta P_E e^{-W_b/L_p}}{e^{W_b/L_p} - e^{-W_b/L_p}}$$

These parameters are applied to eqn no. (3) which give full expression for excess hole distribution in the base region.

$$\delta p(x_n) = \frac{(\Delta P_C - \Delta P_E e^{-W_b/L_p}) e^{x_n/L_p} + (\Delta P_E e^{W_b/L_p} - \Delta P_C) e^{-x_n/L_p}}{e^{W_b/L_p} - e^{-W_b/L_p}}$$

if we assume that the collector j_n is strongly RB i.e. $\Delta P_C \cong P_{n0}$ and eqbm hole conc'n P_{n0} is negligible compared with the injected conc'n ΔP_E { i.e. $P_{n0} \cong 0 \Rightarrow \Delta P_C \cong 0$ }
 $(\Delta P_E \gg P_{n0})$

$$S_p(x_n) = \frac{-\Delta P_E e^{-W_b/L_p} e^{x_n/L_p} + \Delta P_E e^{W_b/L_p} e^{-x_n/L_p}}{e^{W_b/L_p} - e^{-W_b/L_p}}$$

$$= \frac{\Delta P_E (e^{W_b/L_p} e^{-x_n/L_p} - e^{-W_b/L_p} e^{x_n/L_p})}{e^{W_b/L_p} - e^{-W_b/L_p}} \quad \text{--- (4)}$$

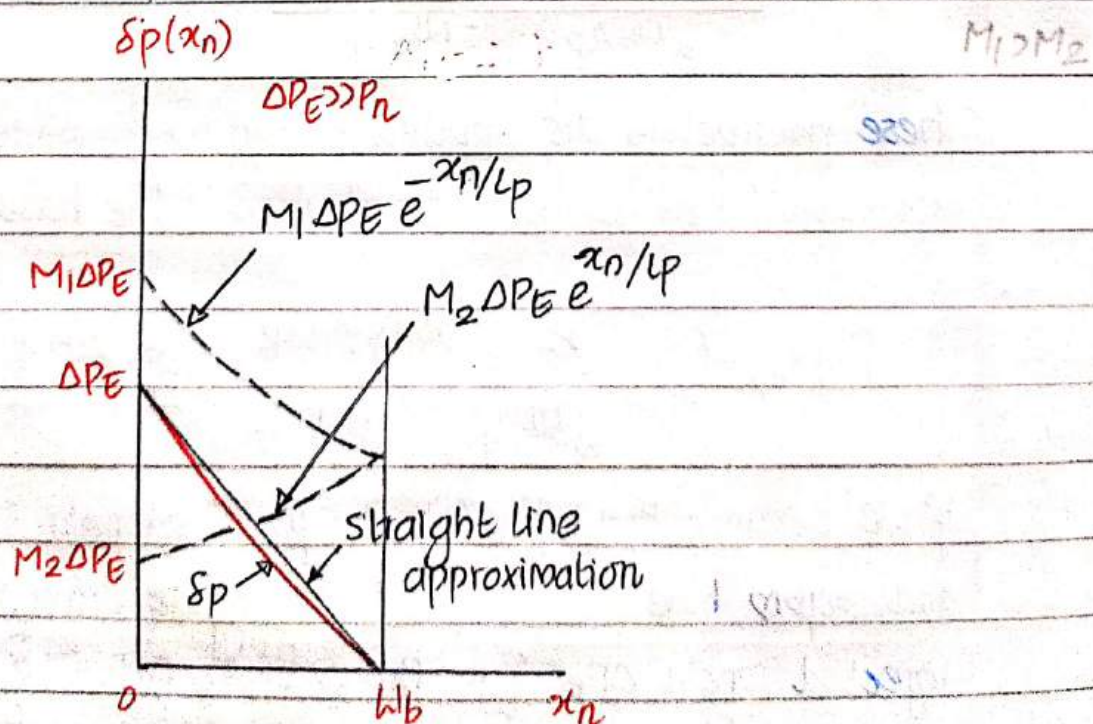
We can re-write eqn (4) as exponential decrease

$$S_p(x_n) = M_1 \Delta P_E e^{-x_n/L_p} - M_2 \Delta P_E e^{x_n/L_p}$$

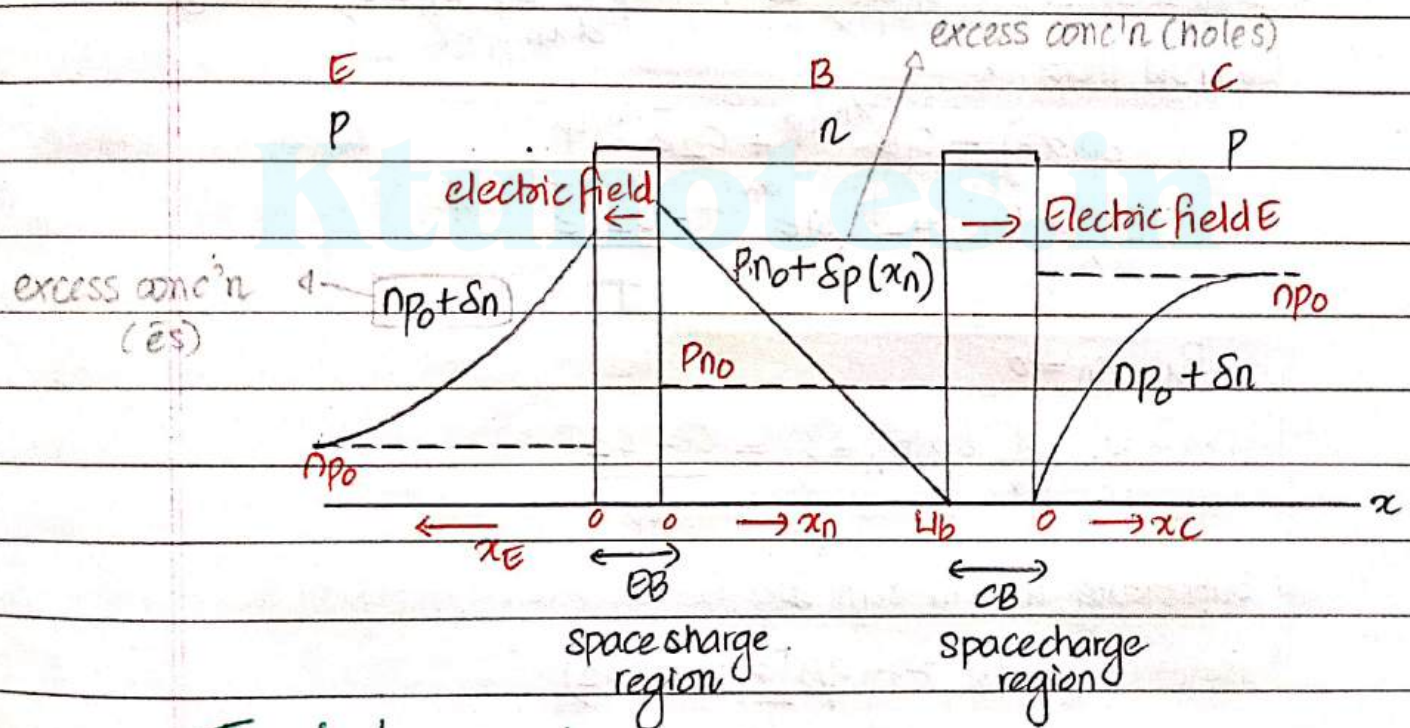
$$\text{where } M_1 = \frac{e^{W_b/L_p}}{e^{W_b/L_p} - e^{-W_b/L_p}}$$

$$M_2 = \frac{e^{-W_b/L_p}}{e^{W_b/L_p} - e^{-W_b/L_p}}$$

The various terms in eqn (4) can be sketched as follows:-



- The various terms in eqn (4) are sketched here and the corresponding excess hole distribution in base region is demonstrated for moderate value of W_b/l_p .
- The $\delta p(x_n)$ varies almost linearly b/w the emitter and collector in depletion region.
- The slight deviation from linearity of distribution indicates the small value of I_B caused by recombination in the base region.
- The $\bar{e}$ distribution in emitter and collector is shown below along with hole distribution in the base.



Terminal currents:

- Evaluating the E and C currents from the gradient of the hole conc'n at each depletion region.
- From the diffusion current derivation,

$$J_p(\text{diff}) = -q D_p \frac{d p(x)}{dx}$$

We have

$$I_p(x_n) = -qAD_p \frac{d}{dx_n} \delta p(x_n)$$

$I_p(x_n=0)$ will give the hole component of I_E .

$I_p(x_n=W_b)$ will give the hole component of I_C .

The hole component of I_E is represented as I_{Ep} .

For $\gamma=1$

$$I_{Ep} \approx I_E$$

$$\therefore I_{Ep} \approx I_E = I_p(x_n=0) = -qAD_p \frac{d}{dx_n} \delta p(x_n=0) \quad \text{--- ①}$$

We know that

$$\delta p(x_n) = c_1 e^{x_n/L_p} + c_2 e^{-x_n/L_p}$$

$$\therefore \frac{d}{dx_n} \delta p(x_n) = \frac{1}{L_p} c_1 e^{x_n/L_p} - \frac{c_2}{L_p} e^{-x_n/L_p}$$

Put $x_n=0$

$$\therefore \frac{d}{dx_n} \delta p(x_n=0) = \frac{c_1 - c_2}{L_p}$$

Substitute this in eqn ①

$$\therefore I_E = -qAD_p \frac{(c_1 - c_2)}{L_p}$$

$$= qAD_p \frac{(c_2 - c_1)}{L_p} \quad \text{--- ②}$$

||| If we neglect the e^- 's crossing from $C \rightarrow B$ in the collector reverse saturation current, I_C is made up entirely of holes

entering the collector depletion region from the base.

$\therefore I_p(x_n = W_b)$ gives the collector current I_c .

ie $I_c = I_p(x_n = W_b)$

$$= q A D_p \left[c_2 e^{-W_b/L_p} - c_1 e^{W_b/L_p} \right]$$

Substitute the value of c_1 and c_2 in I_c and I_E .

$$\therefore I_E = q A D_p \left\{ \left[\frac{\Delta P_E e^{W_b/L_p} - \Delta P_C}{e^{W_b/L_p} - e^{-W_b/L_p}} \right] - \left[\frac{\Delta P_C - \Delta P_E e^{-W_b/L_p}}{e^{W_b/L_p} - e^{-W_b/L_p}} \right] \right\}$$

$$= q A D_p \left[\frac{\Delta P_E (e^{W_b/L_p} + e^{-W_b/L_p}) - 2\Delta P_C}{e^{W_b/L_p} - e^{-W_b/L_p}} \right]$$

We know that

$$\frac{e^x + e^{-x}}{2} = \cosh(x) \text{ and } \frac{e^x - e^{-x}}{2} = \sinh(x)$$

$$\Rightarrow e^x + e^{-x} = 2\cosh(x) \text{ and } e^x - e^{-x} = 2\sinh(x)$$

$$\therefore I_E = q A D_p \left[\frac{2\Delta P_E \cosh(W_b/L_p) - 2\Delta P_C}{2\sinh(W_b/L_p)} \right]$$

$$\Rightarrow I_E = q A D_p \left[\Delta P_E \coth(W_b/L_p) - \Delta P_C \operatorname{cosech}\left(\frac{W_b}{L_p}\right) \right]$$

$$\text{Similarly } I_c = q A D_p \left[\Delta P_E \operatorname{cosech}\left(\frac{W_b}{L_p}\right) - \Delta P_C \coth(W_b/L_p) \right]$$

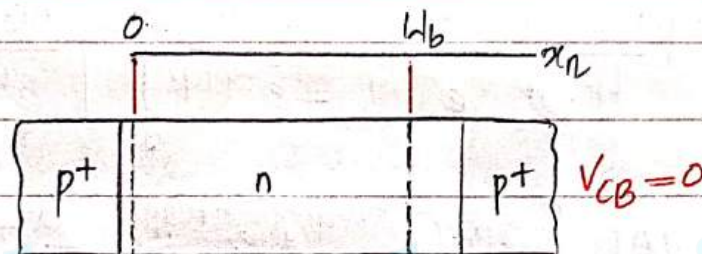
$$\cosh(x) - \operatorname{cosech}(x) = \tanh\left(\frac{x}{2}\right)$$

$$I_B = I_E - I_C$$

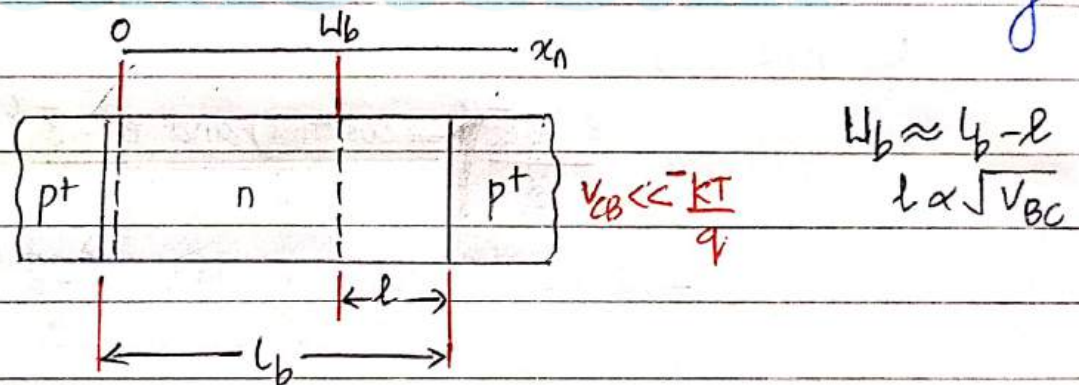
$$= qAD_p \left[(\Delta P_E + \Delta P_C) \left(\cosh\left(\frac{W_b}{L_p}\right) - \operatorname{cosech}\left(\frac{W_b}{L_p}\right) \right) \right]$$

$$I_B = qAD_p \left[(\Delta P_E + \Delta P_C) \tanh\left(\frac{W_b}{2L_p}\right) \right]$$

BASE NARROWING / BASE WIDTH MODULATION / EARLY EFFECT



Base narrowing

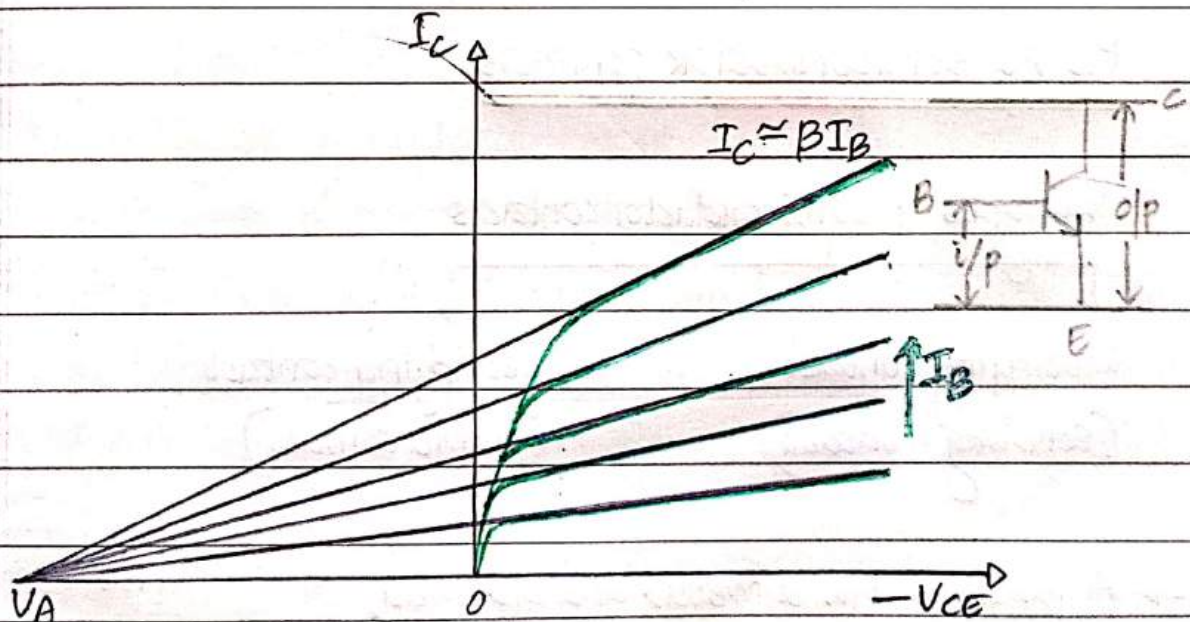


- In case of transistors, so far we have assumed that the effective base width W_b is independent of the bias voltages applied to collector and emitter junctions. The assumption is not always valid.
- The above fig shows the decrease in base width as reverse bias on collector jn is increased.
- If the base region is lightly doped, the depletion region at the reverse biased collector jn can extend significantly into the

n-type base region.

- As collector voltage is increased, the space charge layer takes up more of the metallurgical width of the base L_b and as a result the effective base width W_b is decreased.
- The effects of base narrowing can be seen in collector characteristics for common-emitter configuration.
- The decrease in W_b causes β to increase. As a result the collector current I_c increases with collector voltage rather than staying constant as predicted from the simple treatment.
- The slope introduced by the Early effect is almost linear with I_c and common-emitter characteristics extrapolate to an intersection with voltage axis at V_A called the **Early voltage**.

$\beta = \frac{I_c}{I_b}$
base to
collector
current
amplification
factor



We know from previous chapter that

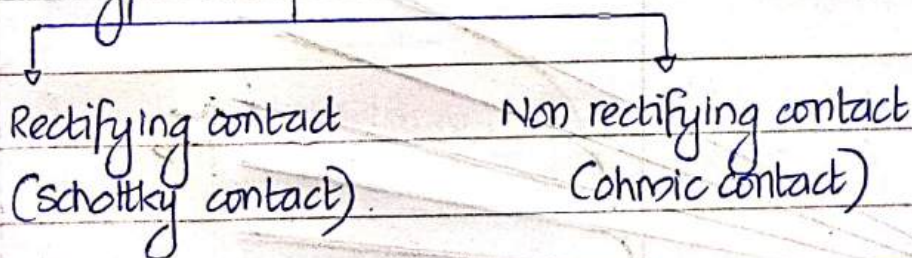
$$x_{n0} = \sqrt{\frac{2\epsilon V_0}{q} \left[\frac{N_a}{N_d(N_a + N_d)} \right]}$$

Replacing V_0 with $V_0 - V_{CB}$ and V_{CB} is taken large and -ve; We can approximate the length l of collector junction depletion region in n material as
$$l = \sqrt{\frac{2\epsilon V_{CB}}{qN_D}}$$

- If the reverse bias on collector junction is increased large enough, it is possible to decrease W_b to the extent that the collector depletion region essentially fills the entire base. This is known as **punch through**.
- In this punch through condition, holes are swept directly from emitter to collector and transistor action is lost.
- Punch through is a breakdown effect that is generally avoided in circuit designing.

METAL SEMICONDUCTOR CONTACTS

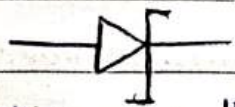
Two types of semiconductor contacts



- A junction b/w a metal and slc may behave like a diode or it may be an ohmic contact which ~~contact~~ conduct well in both directions.
- A rectifying metal slc contact is called a schottky diode.

metal	semiconductor
-------	---------------

- If the contact acts as a diode - schottky (rectifying) contact
 If it creates a movt of charge carres (current) in both the directions -
 Non-rectifying / ohmic (no diode action; a simple metal contact to a s/c).

 : schottky diode symbol.

- Schottky → It does not depend on the type of semi conductor (whether it is an n-type / p-type), but on a factor known as ϕ .

- ϕ is a material-specific quantity represented in volts (V).

$\phi_m \rightarrow \phi$ of metal; depends on the metal and $\phi_s \rightarrow \phi$ of s/c; depends on the type of s/c.

- $q\phi$ is called work function, an energy quantity represented in Joules.

Note:- An ideal metal s/c contact is one in which the interface is free from any charge or defects.

- The behaviour of an ideal metal s/c contact depends on relative work function of metal and s/c ($q\phi_m$ and $q\phi_s$)

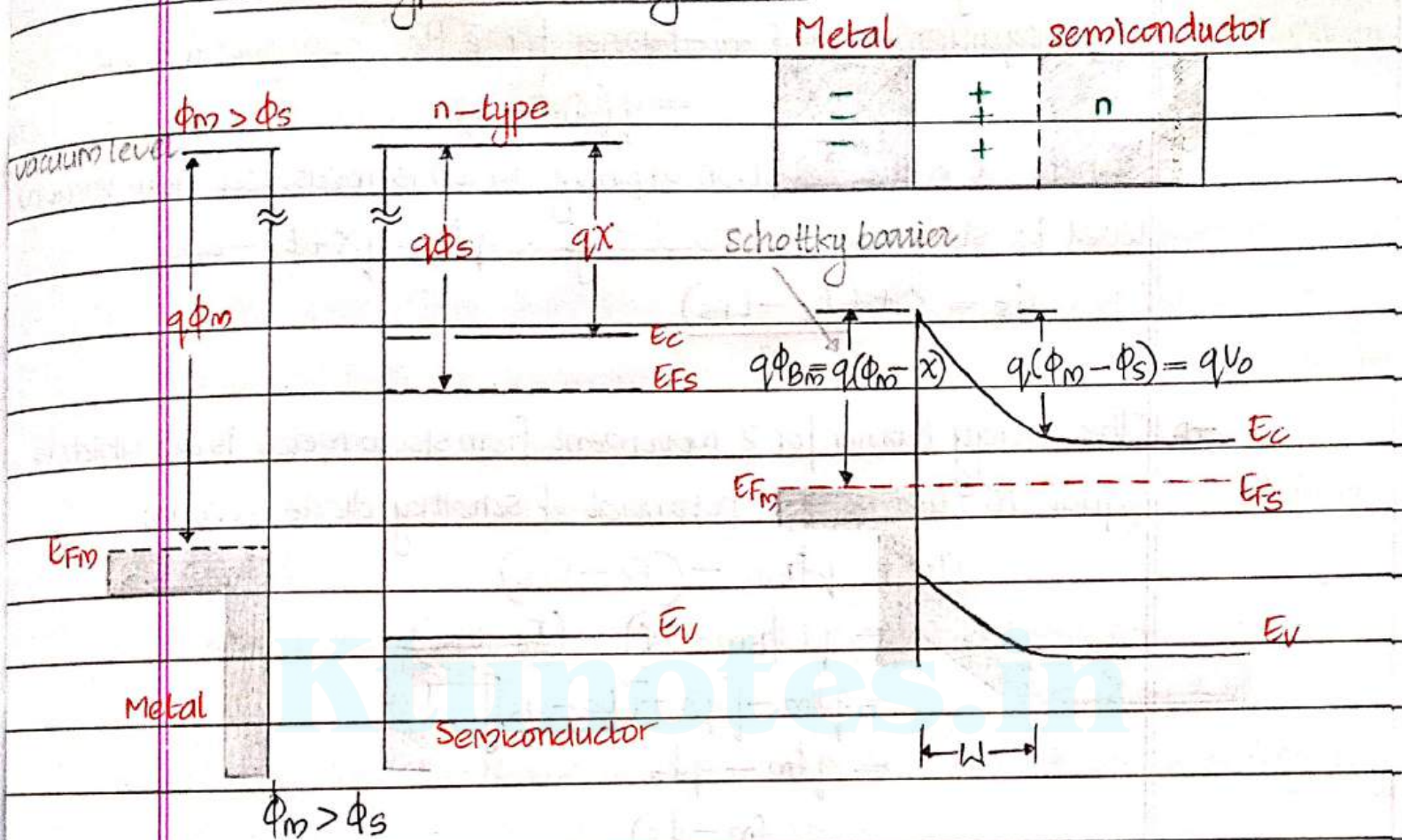
- Work function is the energy required to remove an electron from Fermi level to vacuum outside the material. So, the work function is the energy difference b/w vacuum level and Fermi level. Vacuum level is continuous.

Relative Work Functions

Type of contact	$\phi_m > \phi_s$	$\phi_m < \phi_s$
metal - n type s/c	Rectifying	ohmic
metal - p type s/c	ohmic	Rectifying

For Ref:- Tips to draw Energy-band diagram.

- Draw the energy bands of isolated metal and s/c with common vacuum level.
- Under thermal eqbm (When contact is made b/w metal and s/c) the Fermi levels align so that it is a common horizontal line.
- The Fermi level in metal is taken as reference.
- The location of E_C , E_i and E_V at the reference remains the same as that in isolated energy band diagram w/ metal Fermi level.
- The energy remains flat in neutral region.
- Fermi level in s/c side move up if ~~the~~ -ve potential is applied to s/c and move down if the potential is applied to s/c w/ metal.

RECTIFYING CONTACT :1. metal - n type slc Schottky contact

Band diagram of metal and
slc before joining

Energy band diagram
after joining.

$q\chi$: electron affinity $\rightarrow$ The work function from vacuum to the conduction band minimum value.

- $\rightarrow$ When a contact is made b/w metal and n type slc with $\phi_m > \phi_s$ the energy levels in slc is above that of the metal.
- $\rightarrow$ The average electron energy in slc is more than in metal, that in the metal, so the e^- 's move from slc to metal until the Fermi levels align and equilibrium condition is attained.

→ As $\bar{e}$ move away from n-type slc, the slc near the metal gets depleted of mobile charge carriers.

→ Under equilibrium, $q\phi_{Bm}$ - barrier for $\bar{e}$ flow from metal to slc
 $= q(\phi_m - \chi)$

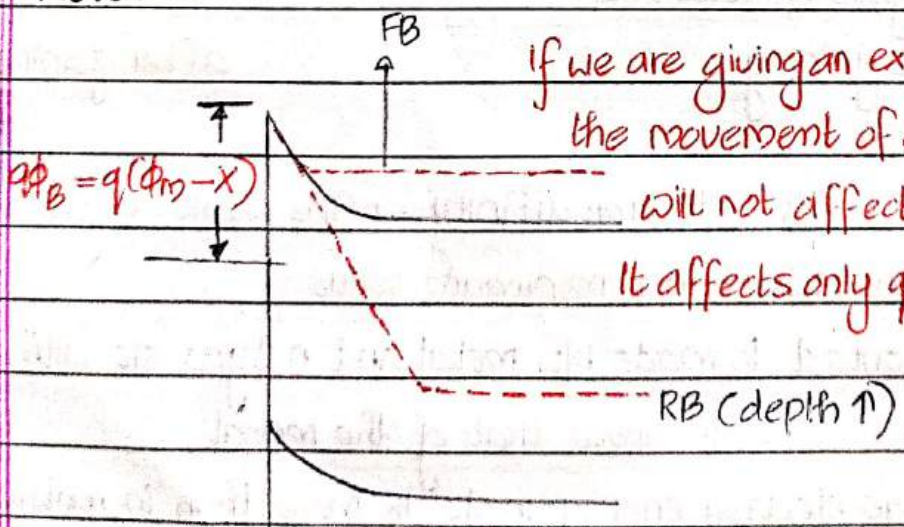
Where χ is the "electron affinity" which is measured from vacuum level to slc conduction band edge. $q\phi_s = q\chi + E_c - E_{Fs}$

$$\phi_s = \chi + \frac{(E_c - E_{Fs})}{q}$$

→ The energy barrier for $\bar{e}$ movement from slc to metal is qV_0 which is equal to the built in potential of Schottky diode given by

$$\begin{aligned} qV_0 &= q\phi_{Bm} - (E_c - E_{Fs}) \\ &= q(\phi_m - \chi) - (E_c - E_{Fs}) \\ &= q\phi_m - [q\chi + (E_c - E_{Fs})] \\ &= q\phi_m - q\phi_s \\ &= q(\phi_m - \phi_s) \end{aligned}$$

Note:-



If we are giving an external biasing for the movement of charge carriers, it will not affect the height, $q\phi_B$. It affects only $q(\phi_m - \phi_s) = qV_0$

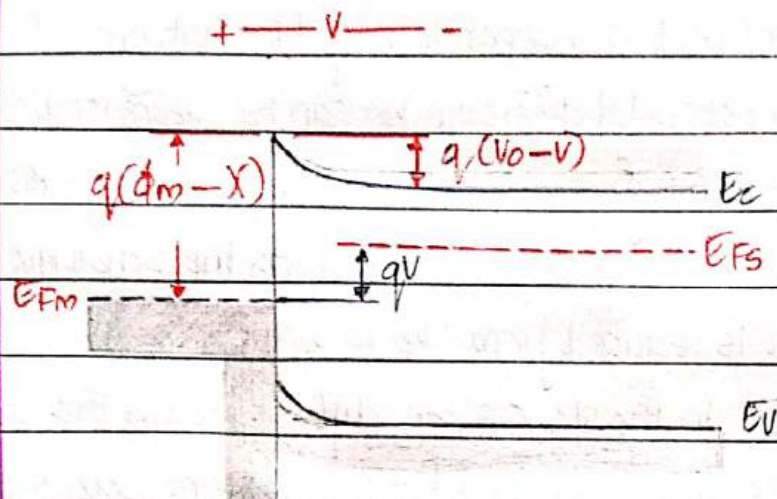
Explanation:- The energy barrier from metal to slc is called

"Schottky barrier" and is independent of bias voltage.

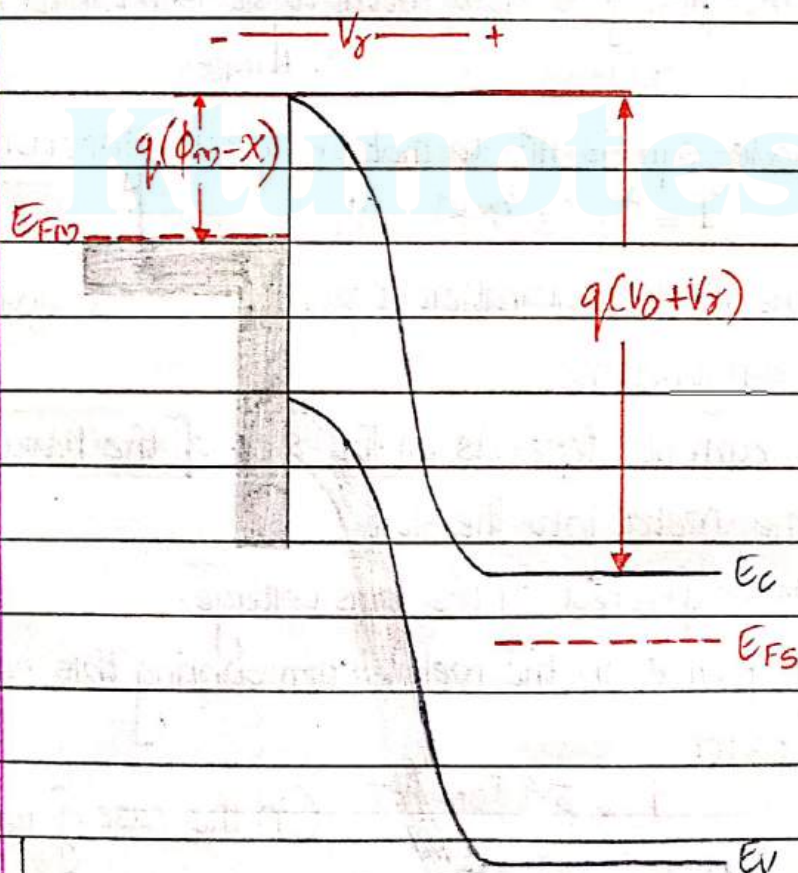
- The equilibrium potential difference V_0 can be decreased or increased by the application of either forward or reverse bias voltage as in p-n junction.
- Let a forward bias voltage V_F be applied to the Schottky barrier; The contact potential is reduced from V_0 to $V_0 - V_F$.
- As a result, the e^- 's in the slc C.B can diffuse across the depletion region to the metal. This gives rise to a forward current across the junction.
- A reverse bias V_R increases the barrier to $V_0 + V_R$ and e^- flow from slc to metal become negligible.
- In either case, the flow of e^- from metal to slc is retarded by the barrier $\phi_m - X$.
- The resulting diode eqn is like to that of the p-n junction.

$$I = I_0 (e^{qV/kT} - 1)$$
- In this case, the reverse saturation current I_0 is not simply derived as it was for the p-n junction.
- The saturation current depends on the size of the barrier ϕ_{BM} for e^- junction from the metal into the slc.
- This barrier is unaffected by the bias voltage.
- The probability of an e^- in the metal surmounting this barrier is given by Boltzmann factor.

$$I_0 \propto e^{-q\phi_{BM}/kT} \quad (\text{in the case of reverse current})$$
- In the case of Schottky contact, the reverse current depends on the barrier height.

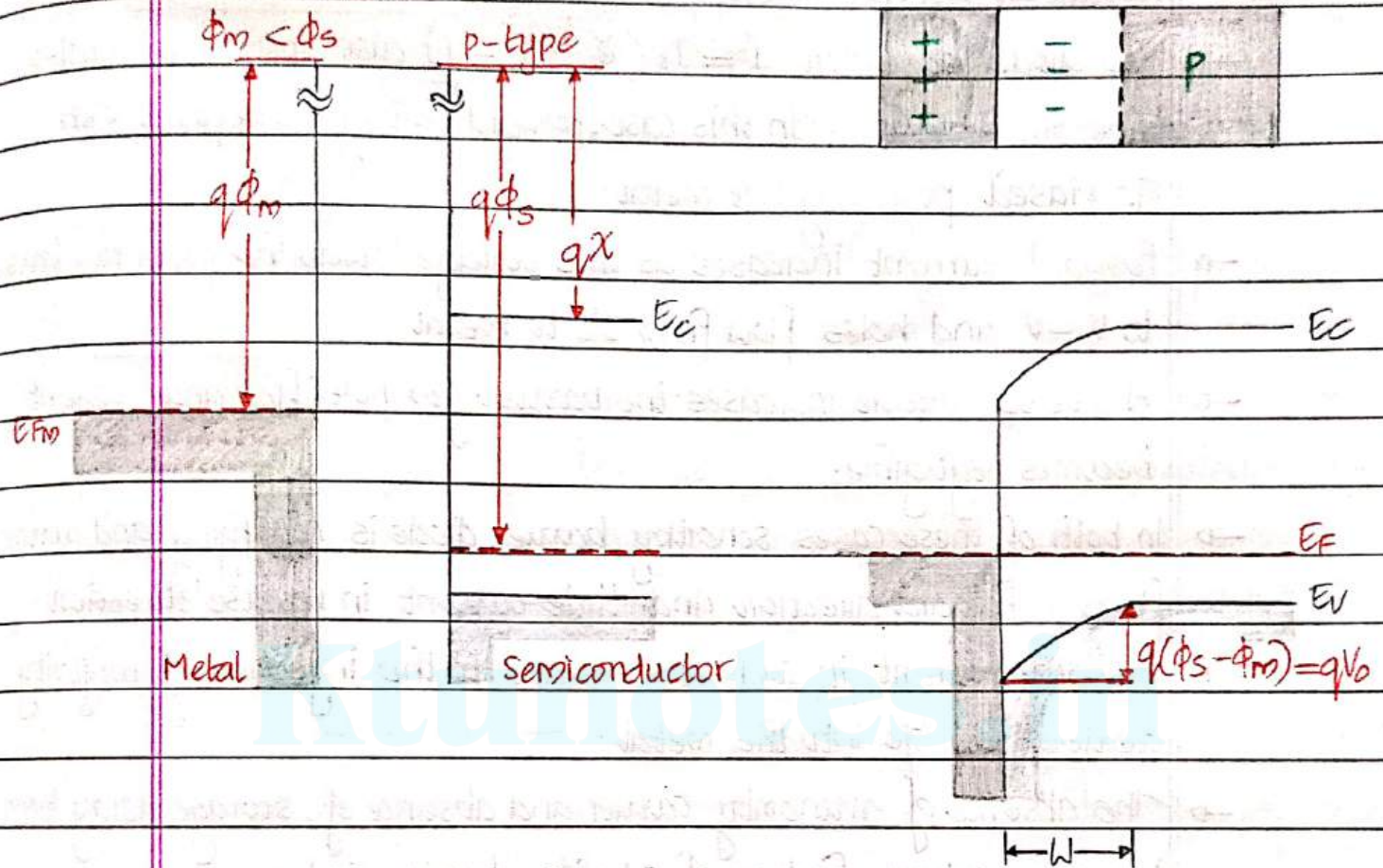


(a) Forward bias



(b) Reverse bias

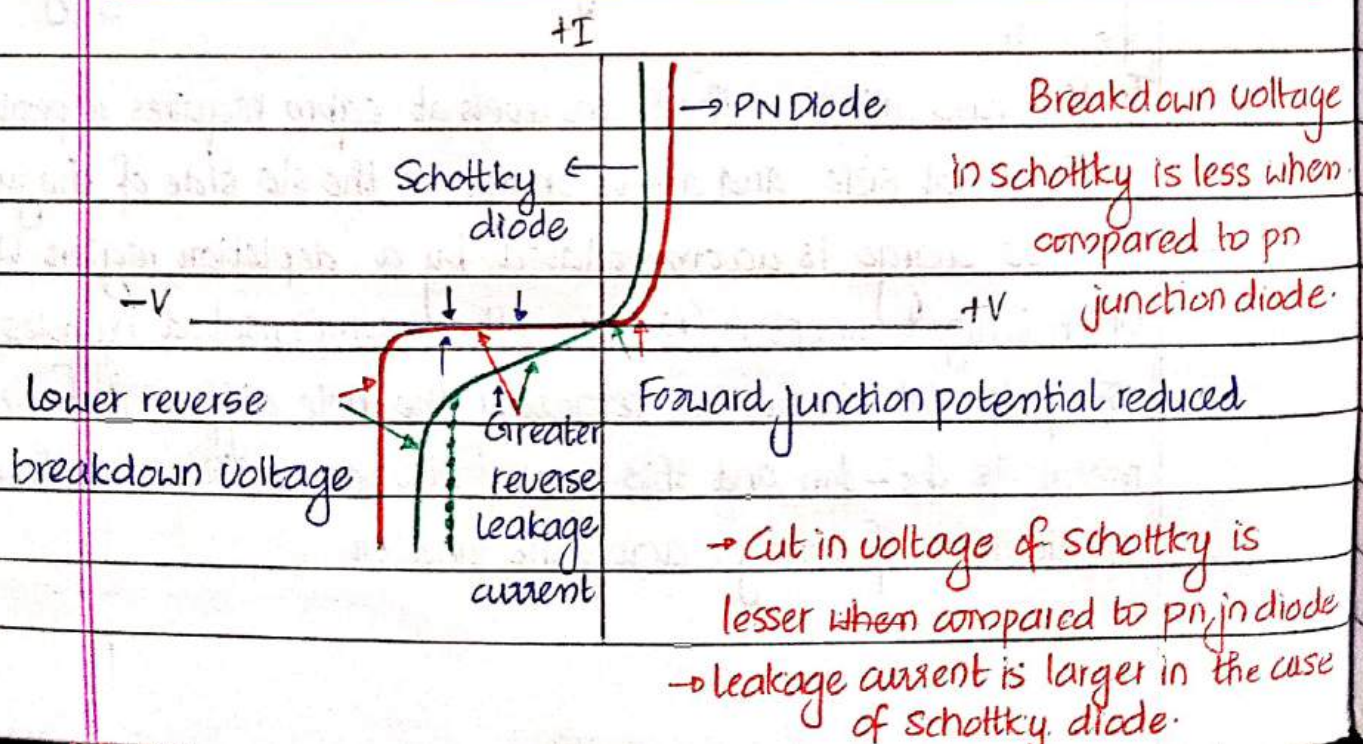
2. Metal-p type slc schottky contact



- A contact b/w metal and p type slc behaves like rectifying contact if $\phi_s > \phi_m$.
- In this case, aligning the Fermi levels at eqbm requires a positive charge on the metal side and a -ve charge on the slc side of the junction.
- The -ve charge is accommodated by a depletion region W_0 in which ionized acceptors (N_A^-) are left uncompensated by holes.
- The potential barrier V_0 , retarding the hole diffusion from slc to metal is $\phi_s - \phi_m$ and this barrier also can be raised or lowered by application of voltage across the junction.

- The electrostatic potential barrier for the charge is opposite to the barrier on electron energy diagram.
- The diode equation $I = I_0(e^{qV/kT} - 1)$ also applies to metal-p type Schottky junction. In this case, forward voltage is defined with Schottky biased positively w.r.t metal.
- Forward current increases as this voltage lowers the potential barrier to $V_0 - V$ and holes flow from Schottky to metal.
- A reverse voltage increases the barrier for hole flow and current becomes negligible.
- In both of these cases Schottky barrier diode is rectifying and current flows in forward direction and little current in reverse direction.
- Forward current in each case is due to the injection of majority carriers from Schottky into the metal.
- The absence of minority carrier and absence of storage delay time is an important feature of Schottky barrier diodes.

COMPARISON OF CHARACTERISTICS OF SCHOTTKY DIODE AND PN DIODE

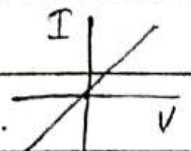


CHARACTERISTIC	SCHOTTKY DIODE	PN JUNCTION DIODE
Forward current mechanism	Majority carrier transport	Due to diffusion currents, i.e. minority carrier transport
Reverse current	Results from majority carriers that overcome the barrier. This is less temperature dependent than for standard PN Junction.	Results from the minority carriers diffusing through the depletion layer. It has a strong temperature dependence.
Turn on voltage	Small—around 0.2V	Comparatively large—around 0.7V.
Switching speed	Fast—as a result of the use of majority carriers because no recombination is required	Limited by the recombination time of the injected minority carriers.

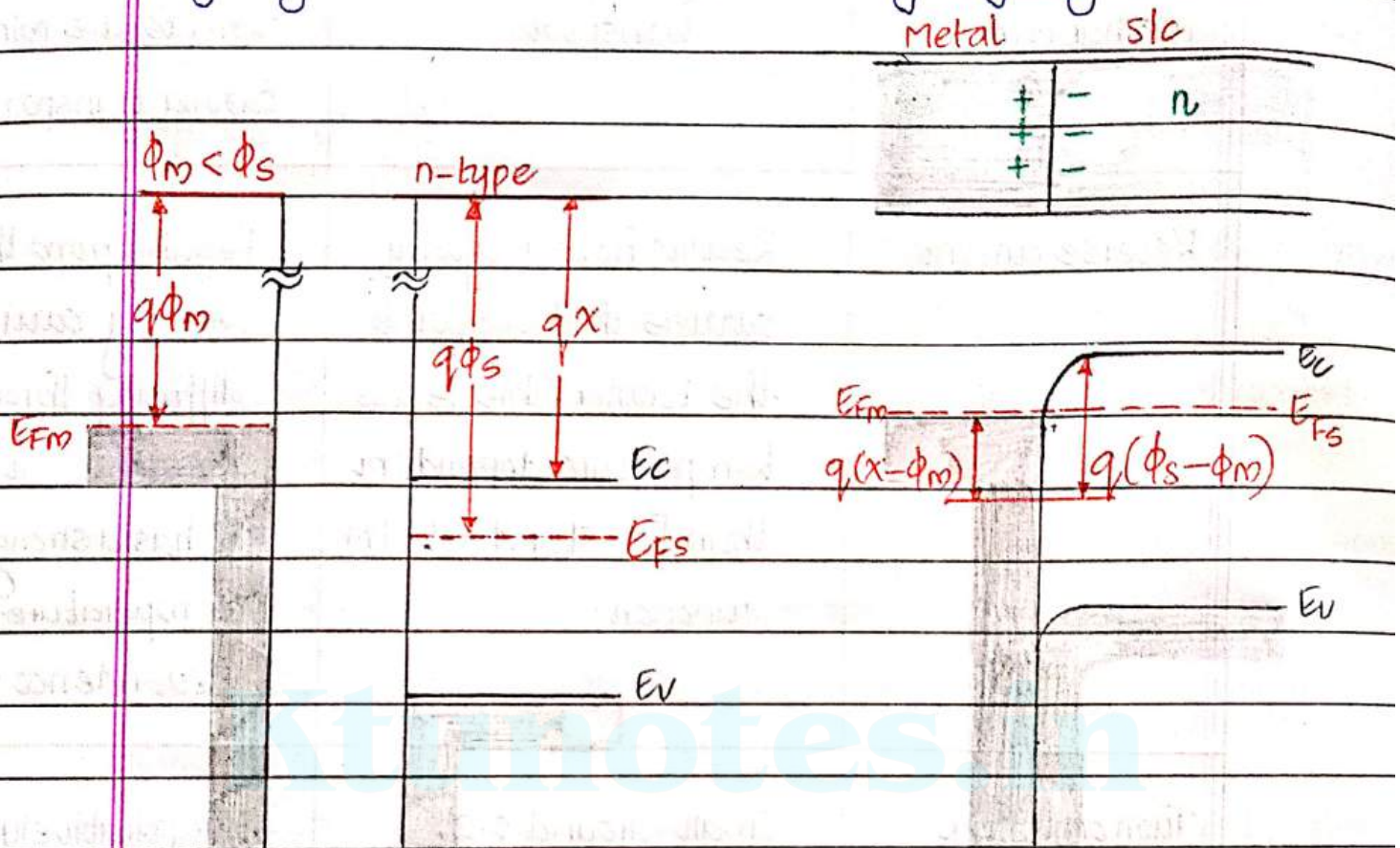
OHMIC CONTACT

1. Metal-n type semiconductor ohmic contact

→ Have linear $I-V$ characteristics in both biasing directions.



- Metal-SiC contacts are ohmic when charge induced in the SiC in aligning the Fermi levels is provided by majority carriers.



→ $\phi_m < \phi_s$

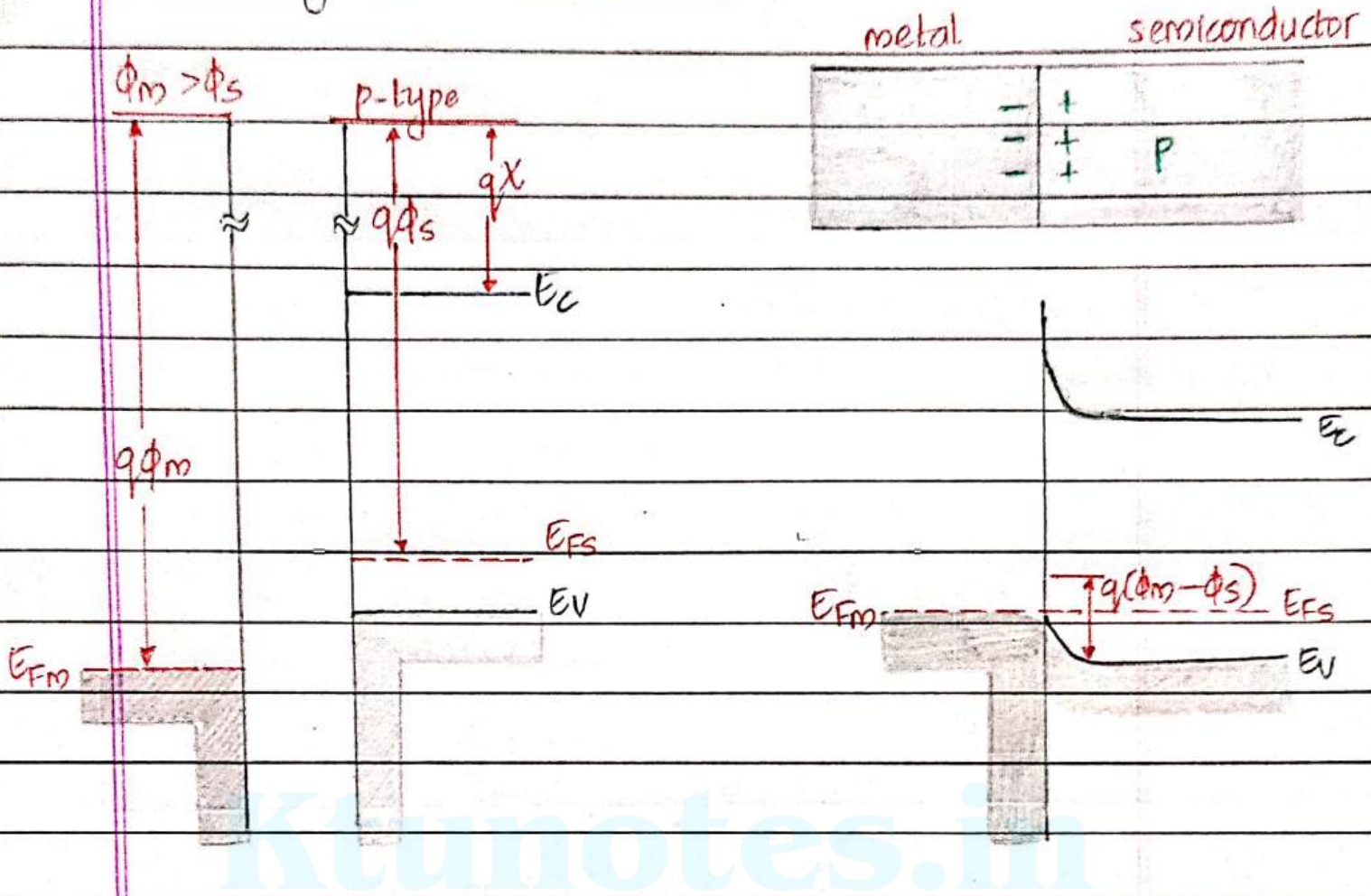
→ Fermi levels are aligned at eqbm by transferring $\bar{e}$'s from metals to SiC.

→ This raises the SiC electron energy (lower electrostatic potential) relative to metal at eqbm.

Note: In metal-SiC contacts become ohmic when the charge induced in the SiC in aligning the Fermi levels is provided by majority carriers.

→ In $\phi_m < \phi_s$ (n-type), the Fermi levels are aligned at eqbm by transferring $\bar{e}$ s from the metal to SiC. This raises the SiC $\bar{e}$ energies at eqbm.

2 → Metal - p type Schottky contact.



- In $\phi_m > \phi_s$ (p-type), hole flow occurs across the junction.
- No depletion occurs in slc in these cases since the $\bar{e}s$ electrostatic potential difference required to align the Fermi levels at eqbro calling accumulation of majority carriers in the slc.
- A practical method for forming ohmic contact is by doping the slc heavily in the contact region. There Thus if a barrier exist at the interface, the depletion width is small enough to allow carriers to tunnel through the barrier (n-type - n^+ layer at interface and p-type - p^+ layer at the interface).

1. DEPLETION LAYER WIDTH	We know that depletion region width, $W = \left[\frac{2\epsilon_s \phi_s}{q N_a} \right]^{1/2}$
2. MAXIMUM DEPLETION LAYER WIDTH (during inversion)	$W = W_m$ during inversion (when $\phi_s = 2\phi_F$) $\therefore W_m = \left[\frac{2 \times 2\epsilon_s \phi_F}{q N_a} \right]^{1/2} \quad \text{--- (2)}$
3. PERMITTIVITY	We know $\epsilon_s = \epsilon_r(Si) \times \epsilon_0$ $= 11.8 \times 8.85 \times 10^{-14} \text{ F/cm}$ (cm in SI)
4.	$\phi_F = \frac{kT}{q} \ln \left(\frac{N_a}{N_i} \right) = 0.0259 \ln \left(\frac{5 \times 10^{15}}{1.5 \times 10^{10}} \right)$ $= 0.329 \text{ V}$
5. FLAT BAND VOLTAGE	$V_{FB} = \phi_{ms} - \frac{Q_i}{C_i}$
6. CAPACITANCE OF INSULATOR (OXIDE)	$C_i = \frac{\epsilon_i}{d} \quad (\text{per unit area})$ $= \frac{\epsilon_0 \epsilon_r(SiO_2)}{d}$
7. THRESHOLD VOLTAGE	$V_T = \phi_{ms} - \frac{Q_i}{C_i} - \frac{Q_d}{C_i} + 2\phi_F \quad \text{---}$
8. CHARGE ON DEPLETION LAYER	$Q_d = -q N_a W_m$ $= -1.6 \times 10^{-19} \times 5 \times 10^{15} \times 4.14 \times 10^{-5}$
9. MINIMUM CAPACITANCE	$C_{min} = \frac{C_i C_{dmin}}{C_i + C_{dmin}} \quad \text{--- (5)}$ $C_{dmin} = \frac{\epsilon_s}{W_m} \quad (\text{per unit area})$

10. MIN CAPACITANCE IN DEPLETION

$$C_{dmin} = \frac{\epsilon_s}{W_n} \quad (\text{per unit area})$$
$$= \frac{\epsilon_0 \epsilon_r (Si)}{W_n}$$

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